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TOWARD AN INTEGRATIVE MACROEVOLUTIONARY APPROACH FOR IMPROVING THE ASSESSMENT OF PLANT LINEAGE VULNERABILITY: THE CASE OF ARACEAE

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*Les travaux sur les Araceae se poursuivent,
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0.2 CONTRIBUTION OF THIS STUDY TO BIODIVERSITY CONSERVATION

La vulnérabilité d'une lignée ne dépend pas uniquement de l'intensité des menaces auxquelles elle est soumise, mais aussi de sa capacité intrinsèque à répondre aux perturbations environnementales, une capacité largement déterminée par son histoire évolutive et les contraintes héritées (Lavergne *et al.*, 2013; Tanentzap *et al.*, 2020). Or, l'évaluation actuelle du risque d'extinction repose principalement sur une vision statique du vivant, fondée sur données démographiques et spatiales de court terme, limitant fortement l'évaluation à quelques espèces bien documentées, sans prise en compte des processus évolutifs, ou des dynamiques d'habitats à long terme (Mace *et al.*, 2008b). Les processus macroévolutifs identifiés dans cette étude, transitions entre niches et aires géographiques, conservatisme de niche et effet des formes de vie sur la colonisation, constituent des indicateurs prometteurs pour pallier cette limite, en permettant d'identifier les lignées intrinsèquement vulnérables par leurs propres dynamiques évolutives, indépendamment des pressions anthropiques directes (Weil *et al.*, 2025; Davies, 2019). Les résultats obtenus suggèrent par exemple que le géophytisme, en verrouillant les lignées dans des niches structurées par les régimes de feu, réduit leur capacité à coloniser de nouvelles niches, les rendant intrinsèquement dépendantes du maintien de ces régimes de perturbation naturels. À l'inverse, l'hémi-épiphytisme, en exposant les espèces à des conditions environnementales contrastées au cours de leur développement et en facilitant les transitions entre niches forestières, pourrait conférer une plus grande résilience face aux changements environnementaux. Ces dynamiques macroévolutives pourraient ainsi servir de proxy pour estimer le risque d'extinction intrinsèque des lignées non encore évaluées, dont seulement 0,87% des espèces de la famille ont à ce jour été évaluées par la Liste rouge de l'UICN (IUCN, 2026).

Au-delà de l'identification des lignées vulnérables, cette approche renseigne sur les processus écologiques naturels à préserver indépendamment d'un habitat ou d'une espèce isolée. La persistance des lignées géophytes dépend du maintien de régimes de feu naturels calibrés, dont tant la suppression totale (par des actions humaines) que l'intensification incontrôlée (par les changements climatiques) constituent des scénarios défavorables ; la résilience potentielle des lignées hémi-épiphytes, liée à des syndrome de traits multipliant les taux de dispersion entre niches, constitue une issue évolutive conditionnée avant tout par le maintien des forêts tropicales et de leur stratification verticale. Dans les deux cas, la conservation des espèces et celle des processus écologiques qui ont façonné leurs stratégies actuelles sont indissociables et sont toutes deux prises en compte dans cette approche.

Appliquée à l'échelle des espèces, et combinée aux taux de diversification estimés à partir de phylogénies datées, cette approche permettrait dans un premier temps de construire un modèle de vulnérabilité intrinsèque, auquel pourraient être intégrés dans un second temps les effets des modifications climatiques prédites et des pressions anthropiques actuelles. Une telle approche permettrait également d'identifier les lacunes taxonomiques au sein de certains clades et de révéler des syndromes de traits associés à des lignées spécifiques, fournissant ainsi des critères objectifs pour la délimitation d'espèces là où les données sont insuffisantes. Ceci est d'autant plus important que la reconnaissance formelle d'un taxon constitue un prérequis indispensable à sa protection juridique et à l'attribution de financements de conservation. Cette approche intégrative offre une voie vers une évaluation plus complète et plus dynamique de la biodiversité, prenant en compte l'ensemble des espèces, des processus et des habitats, et non uniquement les taxons les mieux documentés.

1 SCIENTIFIC STUDY REPORT [6884 words]

1.1 Introduction [1749 words]

Biodiversity is essential to ecosystem functioning and the services they provide, including primary production, climate regulation, and biogeochemical cycles (Loreau *et al.*, 2001). Its ongoing erosion, accelerated by human activities, represents a major challenge due to potentially irreversible losses compromising the stability and resilience of ecological systems (Cardinale *et al.*, 2012; Ceballos *et al.*, 2017). Conservation biology thus aims to limit these losses by preserving the diversity of life at all levels of biological organisation, from genetic diversity within populations to that of ecosystems at the global scale (Noss, 1990). To address these challenges, conservation has been structured around tools for evaluating, comparing, and prioritising extinction risks (Margules *et al.*, 2000; Wilson *et al.*, 2006; Wilson *et al.*, 2007). Species have emerged as the primary units of reference, both for status assessment and management policies (Mace *et al.*, 2008a). This approach has led to standardised frameworks, notably the Red List of the International Union for Conservation of Nature (IUCN), a central tool at the global scale (Mace *et al.*, 2008a; IUCN, 2026). Vulnerability criteria rely primarily on demographic and spatial indicators, including population size, population trends, and the extent of distribution ranges (Mace *et al.*, 2008a), and tools aiming at improving these assessments, notably species distribution (Elith *et al.*, 2009; Peterson *et al.*, 2012; Crisp *et al.*, 2012) and population viability models (Coulson *et al.*, 2001).

Although these approaches have enabled considerable advances in the standardization and efficiency of conservation, they nonetheless depend on a relatively static representation of living organisms (Mace *et al.*, 2008b; Mace *et al.*, 2014). Species are generally considered as stable entities, whose variations are primarily limited to short-term demographic and geographic fluctuations (Mace *et al.*, 2008b). This view is notably reflected by the niche conservatism hypothesis underlying many species distribution models (Peterson *et al.*, 2012), and by the short-term dynamics in population viability models (Coulson *et al.*, 2001). These limitations have led to growing concerns regarding the methods employed, highlighting the absence of explicit integration of genetic and evolutionary processes in conservation tools (Mace *et al.*, 2026; Hoban *et al.*, 2021). In response, new approaches have emerged: the introduction of genetic considerations via probabilistic approaches integrating inbreeding depression and genetic drift (Mace *et al.*, 2026), and the integration of genetic diversity as a complementary indicator (Hoban *et al.*, 2021). Species distribution models have evolved towards the integration of phenotypic plasticity and the ecological dynamics of the niche (Chevin *et al.*, 2010; Urban *et al.*, 2016). Finally, models integrating evolutionary dynamics have emerged with the aim of better representing the responses of living organisms to environmental changes (Hairston Jr *et al.*, 2005; Pelletier *et al.*, 2009). These models have introduced a long-term component into the equation, and growing number of studies are moving towards its integration.

Recent studies have notably demonstrated a relationship between current extinction risk and various macroevolutionary rates estimated from phylogenies, such as extinction, diversification, or niche evolution rates (Davies *et al.*, 2011; Lavergne *et al.*, 2013; Tanentzap *et al.*, 2020; Greenberg *et al.*, 2017). Several works also highlight that high diversification rates can be associated with increased extinction risk in certain plant lineages (Greenberg *et al.*, 2017; Lavergne *et al.*, 2013). Tanentzap *et al.*, 2020 showed, across 509 plant genera, that rapidly diversifying lineages exhibit a higher proportion of threatened species, associated with reduced distribution ranges and narrower ecological niches.

The success of these new approaches rely on the fact that these macroevolutionary processes play a central role in the structure of current biodiversity and in the capacity of lineages to respond to environmental perturbations over the long term (e.g., colonisation, diversification, or adaptive radiations) (Jablonski, 2022; Rolland *et al.*, 2018). Beyond the links between macroevolutionary rates and current vulnerability, some studies also show that macroevolutionary legacies can interact directly with contemporary anthropogenic pressures. Pu *et al.*, 2025 demonstrate the combined influence of the legacy of past interactions with now-extinct megafauna and current human pressures on seed size in Malagasy plant communities. This work constitutes one of the first empirical examples of a direct interaction between macroevolutionary dynamics and contemporary global change, highlighting the value of integrating these processes into conservation strategies.

Integrating macroevolutionary processes into conservation issues requires understanding how and why current lineages occupy their ecological niches, taking into account their evolutionary history and inherited constraints (Antonelli *et al.*, 2015). In this framework, colonisation constitutes a major driver of lineage distribution, relying on dispersal and/or vicariance processes, combined with establishment adaptations in new ecological niches (Antonelli *et al.*, 2015; Weeks *et al.*, 2014). On the one hand, vicariance, resulting from the fragmentation of a geographic area by geographic or climatic barriers and leading to the isolation and subsequent divergence of populations, is often associated with episodes of rapid diversification during major geological or climatic changes (Weeks *et al.*, 2014; Xing *et al.*, 2017; Wen *et al.*, 2014). On the other hand, the long-distance dispersal hypothesis (LDD) has attracted renewed interest in recent years for understanding the mechanisms underlying large-scale colonisations (Nathan, 2006). This hypothesis proposes that evolutionary associations between plant functional strategies and their dispersal vectors (biotic or abiotic) led to adaptations favouring long-distance dispersal (Liu *et al.*, 2024; Onstein *et al.*, 2017). In tropical palms, Onstein *et al.*, 2017 show that dispersal distance is correlated with disperser size: small fruits are dispersed by small frugivores over short distances, while larger fruits are dispersed by megafauna over long distances. Similarly, in Annonaceae, the analysis of functional traits has demonstrated an important role of megafauna in dispersal during the Cenozoic (Onstein *et al.*, 2019). The study of these interactions is thus particularly relevant for understanding the dynamics of biogeographic structuring and the distribution of plant lineages at the macroevolutionary scale (Nathan, 2006; Maurin *et al.*, 2023).

Although certain fruit functional traits, influence the dispersal of lineages towards similar niches, they do not alone explain their capacity to establish in new niches, which requires specific physiological and ecological adaptations (Wiens *et al.*, 2005; Donoghue, 2008). Establishing into an ecologically different habitat appears indeed to be more constraining than colonising a geographically distant but ecologically similar region (Crisp *et al.*, 2012), making niche conservatism a dominant in the majority of lineages' evolutionary history (Wiens *et al.*, 2005). In mimosoid legumes, 95% of speciation events occurred within the same precipitation regime, illustrating how strongly climate constrains the evolutionary trajectory of lineages (Ringelberg *et al.*, 2023). This conservatism nevertheless has a cost: lineages subject to it exhibit a low capacity to respond to rapid environmental changes, which may increase their long-term vulnerability (Wiens *et al.*, 2005).

Conversely, trait innovations can modify these constraints by enabling the exploitation of new niches, thereby accelerating speciation rates and influencing the biogeographic trajectories of lineages (Donoghue *et al.*, 2014; Rabosky, 2017). These innovations can be reproductive in nature, as in columbines (*Aquilegia*), where the evolution of nectary spurs led to a rapid radiation of approximately 70 species in 5 million years

(Whittall *et al.*, 2007), or vegetative and structural. In Combretaceae, the evolution of cage architectures, combining structural defence through branching and spinescence, facilitated the colonisation of environments subject to strong herbivory pressure (Anest *et al.*, 2024). Similarly, the evolution of the geophytic life form, characterised by renewal buds on underground structures such as tubers or rhizomes, constitutes a functional innovation that facilitated the colonisation of seasonal and disturbed environments (Howard *et al.*, 2020). Howard *et al.*, 2020 show, across nearly 13,000 monocot taxa, that geophytes exhibit significantly higher diversification rates than non-geophytes. This selective advantage appears more closely linked to geophytism as a functional syndrome, notably the capacity for underground storage of resources mobilisable during dormancy, than to any particular morphology (Howard *et al.*, 2020). In monocots, which lack cambium and wood, the vegetative adaptations selected may differ substantially from those of dicots, which have been more widely studied in this context.

Araceae Juss., a major group of monocots, constitute a particularly relevant study model with respect to the evolutionary dynamics explaining their diversity and distribution. With approximately 4,815 species distributed across 164 genera, this family is one of the most diversified among monocots and the richest in the order Alismatales (Cusimano *et al.*, 2011; Catalogue of Life, 2026). The family is thought to have appeared in the Early Cretaceous, around 120-128 Ma, during the breakup of Pangaea, with fossil pollen attributable to the Pothoideae-Monstereae clade dating to 110-120 Ma (Nauheimer *et al.*, 2012; Janssen *et al.*, 2004; Friis *et al.*, 2004). The eight currently recognised subfamilies existed before the Cretaceous-Tertiary boundary, approximately 65.5 Ma ago (Nauheimer *et al.*, 2012; Cusimano *et al.*, 2011). Biogeographic analyses suggest a Laurasian origin for the earliest lineages, associated with humid or aquatic environments, followed by a spectacular radiation towards the tropical regions of Africa, South America, Southeast Asia, and Australasia, with today 90% of genera restricted to intertropical zones, including highly diversified and strictly Neotropical genera such as *Anthurium* Schott. (1,393 spp.) and *Philodendron* Schott. (587 spp.) (Mayo *et al.*, 1997; Nauheimer *et al.*, 2012). Nevertheless, the family occupies a wide range of biomes, from arid deserts to cold temperate zones, up to 4,500 m altitude (Mayo *et al.*, 1997; Croat, 2019).

This diversity of ecological niches is accompanied by strong functional diversity (Croat *et al.*, 2020). Araceae indeed present a great diversity of life forms adapted to environmental constraints, including strictly aquatic, free-floating, helophytic, rheophytic, geophytic (rhizomatous or tuberous), primary and secondary hemiepiphytic, and epiphytic species (Figure 1) (Mayo *et al.*, 1997; Croat, 2019). This ecological diversity might have played a major role in the evolutionary and biogeographic trajectories of the family (Mayo *et al.*, 1997; Croat, 2019). Life forms appear notably strongly linked to climatic and hydric constraints (Mayo *et al.*, 1997). Araceae are structurally and physiologically poorly adapted to cold or arid environments, while lineages occupying temperate regions or high altitudes are predominantly geophytic or helophytic (Croat, 2019). Some lineages have evolved towards hemiepiphytic or epiphytic strategies associated with humid tropical forests (Croat, 2019), and geophytism may constitute a key adaptation for the family to seasonality or more constraining climatic conditions (Figure 1) (Mayo *et al.*, 1997). This last life form is moreover particularly represented in certain relatively primitive lineages (Lasioideae) as well as in several derived tribes of Aroideae, suggesting either ancient conservation of this trait or repeated acquisitions throughout the evolutionary history of the family (Mayo *et al.*, 1997). Analysing the processes that have led to ecological and functional diversification in Araceae could contribute to improving the conservation of this group by identifying the most vulnerable lineages in the face of current global changes, in a context where only



Figure 1: Diversity of Araceae life forms. (A) Geophyte, *Biarum olivieri* Blume, ©Ron Frumkin; (B) Terrestrial herb, *Dieffenbachia oerstedii* Schott, ©Dirk Mezger; (C) Primary hemi-epiphyte, *Rhaphidophora korthalsii* Schott, ©cheavy; (D) Secondary hemi-epiphyte, *Philodendron minarum* Engl., ©Jonathan Santos; (E) Epiphyte, *Anthurium gracile* (Rundg) Lindl., ©David Scherberich; (F) Rheophyte, *Bucephalandra* sp., ©Samuel Lee; (G) Helophyte, *Typhonodorum lindleyanum* Schott; (H) Free-floating aquatic, *Pistia stratiotes*, ©Patrick Hanly; (I) Aquatic, *Cryptocoryne striolata*, ©Mohamad Jakaria.

422 Araceae species have been assessed by the Red List, representing barely 0.87% of species in the group, of which 163 are classified in a threatened category (IUCN, 2026).

The present study take place within a framework aiming to (1) identify the main ecological strategies of current Araceae, (2) evaluate their link with colonisation processes, and (3) test whether the macroevolutionary dynamics of lineages can predict their intrinsic vulnerability independently and in association with anthropogenic pressures. The work carried out here will address questions relating to ecological strategies and their relationship with the diversification of occupied ecological niches (1 and 2), in order to discuss their potential implications for the conservation of Araceae and establish the foundations of the model that can be developed over the longer term (3).

For the present study, we formulate the following hypotheses. Concerning the diversity and distribution of ecological niches (1), we suppose that: (1.i) although Araceae are predominantly associated with humid tropical environments, they occupy an important diversity of ecological niches within these environments; and (1.ii) the occupied niches are associated with functional traits linked to life form. Regarding colonisation mechanisms (2), we suppose that establishment in new ecological niches has been facilitated by certain traits linked to life form, notably in geophytes, enabling persistence in seasonal environments, independently of historical colonisation processes such as dispersal or vicariance.

1.2 Materials and Methods [1756 words]

Author Contributions

All analyses were conducted by me (the student). The spatial data were extracted from GBIF, the trait data and phylogeny used from the authors cited below.

Collaborative Project Background

This study aims to build a PhD projet aiming at developing a predictive model of global lineage vulnerability within the Araceae family to improve their conservation. The objective is to understand past macroevolutionary and ecological processes to predict future outcomes in the context of climate change. Within this framework, Daniel Dutra will publish the first species-level phylogeny of this scale and resolution (Angiosperm353) of the Araceae in the coming weeks (manuscript in revision). The data and methodological choices result from exchanges with Araceae specialists: Simon Mayo, Anna Haigh, Daniel Dutra, Marc Gibernau, and Monica Carlsen.

Github

All data (except rasters, TIFF files, and other large files or those not permitted for distribution), figures, and scripts are available on GitHub, along with a Shiny application (Native_Range.r) that allows interactive visualization of species and genus occurrences and native ranges of the Araceae family on a map, including a buffer zone. GitHub link :

https://github.com/EliauVuillemot/Araceae-GitHub2026/tree/ea515962e0ba75b4c05456c8bec101f568f5941a/Output_M%C3%A9moire/Outputs_aov_phylo

1. Data preparation

Data acquisition

All analyses were conducted in RStudio v2026.01.2. Georeferenced occurrence data for Araceae were downloaded from the GBIF API using the *rgbif* package (Chamberlain *et al.*, 2017). For each species, up to 10 000 georeferenced occurrences were retrieved the accepted scientific name, country code (ISO 3166-1 alpha-2), decimal latitude, and decimal longitude. Country codes were converted to full country names using the *countrycode* package (Arel-Bundock *et al.*, 2018). Strict duplicates were removed.

Restriction to native distribution ranges

To assess the effect of traits potentially facilitating niche colonization across lineages, occurrences corresponding to introduced or cultivated individuals outside native ranges were excluded. Each occurrence dataset was spatially cross-referenced with species native distributions derived from the World checklist of vascular plants (WCVP), using the *rWCVP* package (Brown *et al.*, 2023). Occurrences located more than 20 km outside native range boundaries were excluded to minimize errors associated with geometric imprecision of WCVP polygons at range margins.

Geographical data cleaning

Coordinates were processed using *CoordinateCleaner* (Zizka *et al.*, 2019), excluding records with missing or zero coordinates, duplicates, marine occurrences, and records located near country centroids, administrative centroids, or biodiversity institutions. Species represented by fewer than five occurrences after cleaning were excluded. In total, 1,981 species across 110 genera were retained, representing 338,611 occurrences.

2. Extraction and aggregation of environmental variables

Variables selection

The 19 WorldClim bioclimatic variables (version 2.1, 2.5 arc-min resolution) and elevation were extracted for each occurrence using the *terra* and *geodata* packages (Hijmans *et al.*, 2020; Hijmans, 2021). Occurrences with missing values for at least one bioclimatic variable were excluded. Occurrence points were then rasterized at 5 arc-min resolution by rounding down coordinates. Four additional environmental variables were extracted via spatial overlay on the same 5 arc-min grid. The annual aridity index (ratio between annual precipitation and potential evapotranspiration) (Zomer *et al.*, 2022) was used as a proxy for hydric stress. Soil cation exchange capacity (CEC) was obtained from the SoilGrids portal (soilgrids.org), and the number of dry months (NbDryM) was derived from preprocessed global raster layers resampled to the target resolution (Maurin *et al.*, 2023). Given the sensitivity of Araceae to water availability, three hydrological variables at 5 arc-min resolution (water coverage, water renewal, and hydrological seasonality) were extracted using a point-to-tile intersection approach (Pekel *et al.*, 2016). For each raster file, occurrences falling within the spatial extent of each tile were identified, and corresponding pixel values were extracted. Distance to coastline was computed using the *distancetocoast* package (McCloy *et al.*, 2023).

Grid-cell and species-level aggregation

Occurrences were spatially thinned by retaining a single record per species per 5 arc-min grid cell. All environmental variables were subsequently averaged across all occupied grid cells for each species, producing a species-level synthetic environmental niche vector used in downstream comparative analyses.

3. Ecological niche characterisation

Environmental variable preprocessing

Ecological niches were defined using filtered environmental data. Environmental variables were standardized using decostand (method = "norm", margin = 2) to ensure comparable contributions of each variable to multivariate analyses. Variable selection combined statistical criteria and biological relevance. Variables previously identified in the literature as potential drivers of differentiation among groups were retained (Mayo *et al.*, 1997; Croat, 2019). Among these, only weakly correlated variables ($R^2 < 0.8$) were retained to reduce multicollinearity. The final set of variables included: mean annual temperature (bio1), maximum temperature of the warmest month (bio5), precipitation of the driest quarter (bio17), elevation (alt), number of dry months (NbDryM), and soil cation exchange capacity (CEC).

Multivariate analyses

Species were first compared using an unsupervised hierarchical classification, in order to group individuals sharing similar ecological values across the retained variables (Appendix 3). Pairwise comparisons between species were then computed using Euclidean distance. A hierarchical agglomerative clustering (HAC) was performed on this distance matrix using the hclust function from the stats package (Kartikeya Bolar, 2019), based on dissimilarities calculated with the daisy function from the cluster package (Maechler *et al.*, 1999). The most appropriate linkage criterion was selected by comparing observed and projected distances using the Euclidean norm index, and by assessing agreement with a threshold value to best represent observed dissimilarities (Appendix 2) (Mérigot *et al.*, 2010). Results are presented as a dendrogram, and branches were cut to define coherent clusters. Cluster validity was assessed using the Calinski–Harabasz index, computed with the intCriteria function from the clusterCrit package (Appendix 1) (Desgraupes, 2012). A principal component analysis (PCA) was performed using the ade4 package (Dray *et al.*, 2007) to visualize species distributions in a reduced-dimensional space and facilitate identification of clusters or ecological gradients. Eigenvalues were calculated to quantify the variance explained by each axis.

Niche characterization

To interpret and refine the ecological niche clusters, Köppen–Geiger biome centroids (Beck *et al.*, 2018) were projected onto the unconstrained PCA space (Appendix 4). Additional environmental descriptors were then incorporated, including fire regimes derived from the AVHRR-LTDR ESA CCI Fire product (Chuvieco *et al.*, 2020), covering 1982–2018. Annual fire frequency was computed from monthly presence/absence data and subsequently aggregated annually. Tree cover was extracted from the Global Forest Change dataset (version 1.8), defined as canopy closure percentage for vegetation taller than 5 m (Hansen *et al.*, 2013). For these two variables, classes were defined based on the empirical distribution of values (four or three categories depending on the variable).

Herbivore pressure was estimated using relative herbivore proportions for 265 species, primarily folivores, at a spatial resolution of 0.05° . This metric allows assessment of current herbivore distribution and inference of areas potentially supporting high herbivore densities, including regions where relevant species may have gone locally extinct (Anest *et al.*, 2024). Herbivory was categorized into three classes based on species-specific minimum, maximum, and tolerance range, and subsequently projected into a multidimensional ecological space (Appendix 5).

4. Association between traits and ecological niches

To better understand the evolutionary history of traits across genera in the family, we estimated phylogenetic signal and ecological associations, allowing inference of potential evolutionary origins for each trait.

Phylogeny and trait data

The phylogeny of Araceae used here is the most recent and robust reconstruction currently available for the family (Haigh *et al.*, 2023), built at the genus level. This taxonomic resolution was retained to identify broad evolutionary and structural patterns within the family. The phylogeny is based on a sampling of 128 species representing 111 Araceae genera (approximately 78% of recognized genera), complemented by 12 outgroup taxa from Alismatales. It was generated using targeted sequence capture data (Angiosperms353), supplemented with plastid data derived from off-target reads and public sequence databases. In this study, tree was pruned to retain one representative per genus. Traits were obtained from Cusimano *et al.*, 2011 (81 traits) and complemented by a revised and updated dataset compiled by Daniel *et al.* (in review) (3 additional traits), for which we have not yet tested the other variables (see Table 1 in the Appendix for differences). Traits were coded at the genus level, with multiple possible states per genus. Only genera shared between the phylogeny and the ecological niche dataset were retained for further analyses for a total of 93 genus.

Phylogenetic signal of traits

Phylogenetic signal for each binary trait was assessed using D statistic from Fritz *et al.*, 2010, the best fitting method for binary data, computed with the `phylo.d` function from the *caper* package. A value of $D = 0$ indicates phylogenetic conservation consistent with a Brownian motion model, $D < 0$ indicates stronger-than-Brownian phylogenetic clustering (i.e., strong phylogenetic conservatism), and $D = 1$ indicates a random distribution with respect to phylogeny. Significance was evaluated by comparing observed values to a null distribution generated by randomly permuting trait values across the tips of the tree (1000 permutations; P-value P_{val1}).

Trait-environment associations

Associations between each trait and each ecological niche were tested using phylogenetic ANOVA (`aov.phylo`, package *geiger*), following Garland *et al.*, 1993. This approach generates a null distribution of the F-statistic through Brownian motion simulations on the phylogeny (1000 simulations; Wilks test), thereby accounting for phylogenetic non-independence among genera due to shared evolutionary history.

Identification of traits of interest

To identify traits that might be informative regarding major ecological events in the family history, we combined results from phylogenetic signal and trait-environment association previously obtained. This approach allowed us to classify each trait into four categories, following the framework of Anest *et al.*, 2021. This classification distinguishes: (1) traits primarily shaped by developmental constraints, showing strong phylogenetic signal without environmental association (CD); (2) traits reflecting ecological convergence, showing strong environmental association without phylogenetic signal (C); (3) traits potentially involved in adaptive radiations, exhibiting both strong phylogenetic signal and environmental association (KC); and (4) traits with no detectable signal (N). The latter classification allowed us to select a first set of traits that might be informative for the family history and will be retained for further analyses.

5. Ancestral State Reconstruction

Based on these results, the traits exhibiting the strongest phylogenetic signal, the highest number of significant associations with ecological niches, and ecological relevance according to our hypotheses were selected for further analyses using the *Phytools* package (Revell, 2024). For each trait, three Mk-type character evolution models were compared using a likelihood ratio test: the equal rates model (ER), which assumes an identical transition rate among all states; the symmetric model (SYM), which assumes identical forward and backward rates between each pair of states; and the all-rates-different model (ARD), which estimates an independent rate for every possible transition. The selected model was then used to reconstruct the evolutionary history of the traits via stochastic character mapping, implemented through the `make.simmap` function, with 1000 simulations per trait.

6. Ecological diversification and trait–niche relationships

Patterns of ecological diversification and transitions among ecological niches were analysed using the *BioGeoBEARS* package (Matzke, 2013), by replacing classical biogeographic areas with the ecological niche categories defined in the multivariate analyses. Each genus was assigned to one or multiple ecological niches coded as discrete states, allowing niche transitions to be interpreted as processes of dispersal, local extinction, or environment-dependent cladogenesis. The maximum number of simultaneously occupied niche combinations was limited to three due to computational constraints, with the least frequent niches grouped into a residual category (R).

Six null models were compared: *DEC*, *DEC+J*, *DIVALIKE*, *DIVALIKE+J*, *BAYAREALIKE*, and *BAYAREALIKE+J*, each differing in their treatment of cladogenetic events where *DEC* accounts for vicariance and sympatry, *DIVALIKE* emphasizes vicariance, and *BAYAREALIKE* primarily models sympatric processes while excluding vicariance. The +J variants include an additional parameter (j) that models niche shifts associated with founder-event speciation (i.e., jump dispersal). The parameters d (rate of transition to a new niche), e (local niche loss), and j were estimated by maximum likelihood using the `bears_optim_run` function. Models were compared using AICc, and ancestral niche occupancy probabilities were reconstructed at each node to infer evolutionary trajectories within the family. An additional parameter (m2) was included to account for potential multiplicative effects of a trait on the macroevolutionary dispersal parameter (d).

Geographic analyses were also performed, under the same models within a constrained framework including five continental regions : the Americas, Africa, Eurasia, Australia, and India. India was treated as a distinct geographic region under the hypothesis of prolonged isolation prior to its collision with Eurasia.

1.3 Results [1638 words]

1. Ecological niche characterisation

We identified 13 different niches based on clustering (Figure 2.A). The Calinski index reached its maximum value at 16 groups, but where 13 clusters corresponds to an explicit leveling off of the index (Appendix 1). The value of three clusters appeared to represent the most parsimonious partition with respect to the major ecological niches occupied by the family; however, and as the present study aimed to search for a finer-scale niche identification, we selected the number of niches that both 1) bring new contribution on Araceae

ecological characterisation while still 2) producing a good fit according to Calinski criteria; this led us to fix 13 as the best fitting number of ecological niches. Although variables related to water availability were extracted, the associated results were not conclusive and were therefore not considered in the analyses. In addition to representing distinct climatic micro-envelops, all the niches identified vary in 1) forest cover, and so light resource, 2) disturbance regimes including both fire frequency and herbivory pressure (Figure 4).

Strictly forested and fire-free niches encompass the majority of species (Figure 4). Niches A (Lowland tropical moist forest with dry season) and B (Lowland aseasonal equatorial rainforest) correspond to lowland tropical forests with closed canopies and low fire frequency; they mainly differ in exhibiting markedly higher herbivory pressure and nearly twofold greater annual precipitation in niche B (~ 1000 mm) (Figure 4, 2, 3). Niches E (Humid tropical montane forest mid altitude) and I (Humid tropical montane forest high altitude) represent montane counterparts of this profile, characterized by fully forested habitats, near absence of fire, and increasing herbivory with elevation. Niche L (Boreal and cold temperate forest) represents the extreme end of this profile, with closed habitats, absence of herbivory and fire, the lowest mean annual temperature ($\sim 4^{\circ}\text{C}$), and low elevations (< 500 m). Niche D (Temperate deciduous and mixed forest) shares similar characteristics, although herbivory remains more variable.

In contrast, open niches dominated by fire are clearly standing apart. Niche G (Dry tropical savanna with fire regime) represents the most extreme profile, characterized by open habitats, low herbivory, and very high fire frequency (at least one fire every four years) (Figure 4 and 2). Niche M (Arid Mediterranean desert and steppe) exhibits a similar open and arid profile, lacking herbivory but associated with strong fire regimes, corresponding to extreme environments frequently linked to rocky or sandy substrates. Niche C (Humid tropical savanna with seasonal fire regime) occupies an intermediate position, with variable tree cover, herbivory, and fire frequency (Figure 4, 2, 3).

Several niches display mixed or transitional ecological profiles. Niches F (Mediterranean and subtropical dry forest and shrubland), J (Humid tropical montane savanna and grassland), and K (Subtropical high-mountain shrubland and forest) are characterized by semi-open to open habitats with variable herbivory and fire regimes. Niche H (Temperate montane grassland and forest) differs in maintaining relatively closed forest habitats despite variable herbivory and fire, functionally linking it to transitional niches. Notably, niches I, J, and K, corresponding to the highest elevations (up to 3500 m), are primarily distributed along the Himalayan range and are largely represented by the genus *Arisaema*. Among these, niche I is characterized by high soil fertility and humid environmental conditions.

2. Phylogenetic signal and environmental affinity estimations

84 traits (84 (CD) and 42 (KC)) exhibited a strong phylogenetic signal, indicating that at least one state within each variable is phylogenetically conserved (Figure 5). 46 traits (4 (C) and 42 (KC)) showed a significant association (positive or negative) with at least one specific ecological niche. Among these, the four traits lacking phylogenetic signal (C) included stem type, cotyledon type, and leaf traits.

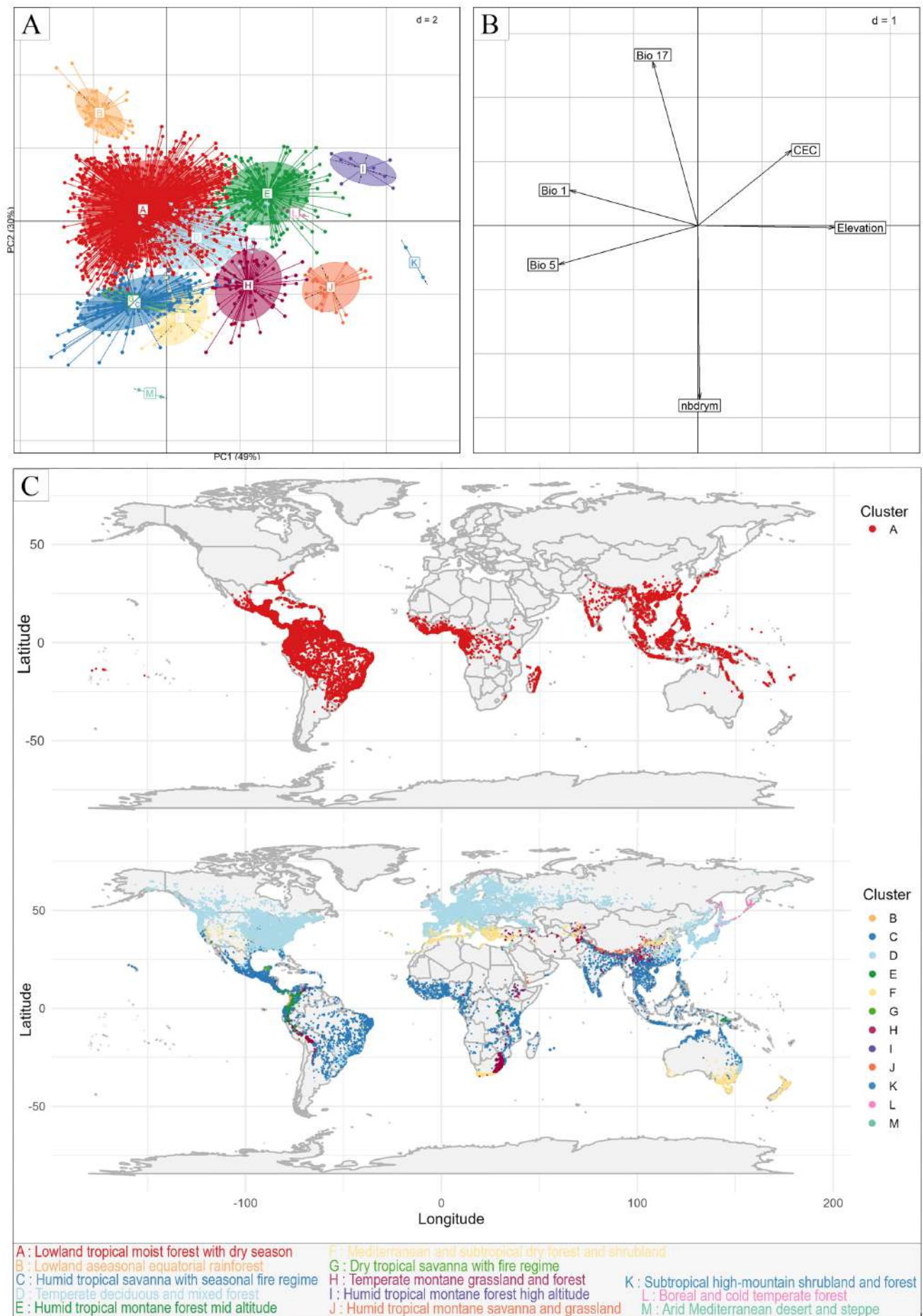


Figure 2: (A) Principal component analysis on the first two principal components (PC1 : 49% ; PC2 : 30%)(PCA) of the mean occurrence values of Araceae species based on six environmental variables: annual mean temperature (Bio1), maximum temperature of the warmest month (Bio5), precipitation of the driest quarter (Bio17), Elevation (Alt), number of dry months (NbDryM), and soil cation exchange capacity (CEC). Colors correspond to the 13 clusters defining the identified ecological niches. (B) Correlation circle of the environmental variables projected onto the first two PCA axes. (C) Geographic distribution of Araceae occurrences retrieved from GBIF, colored according to cluster membership.

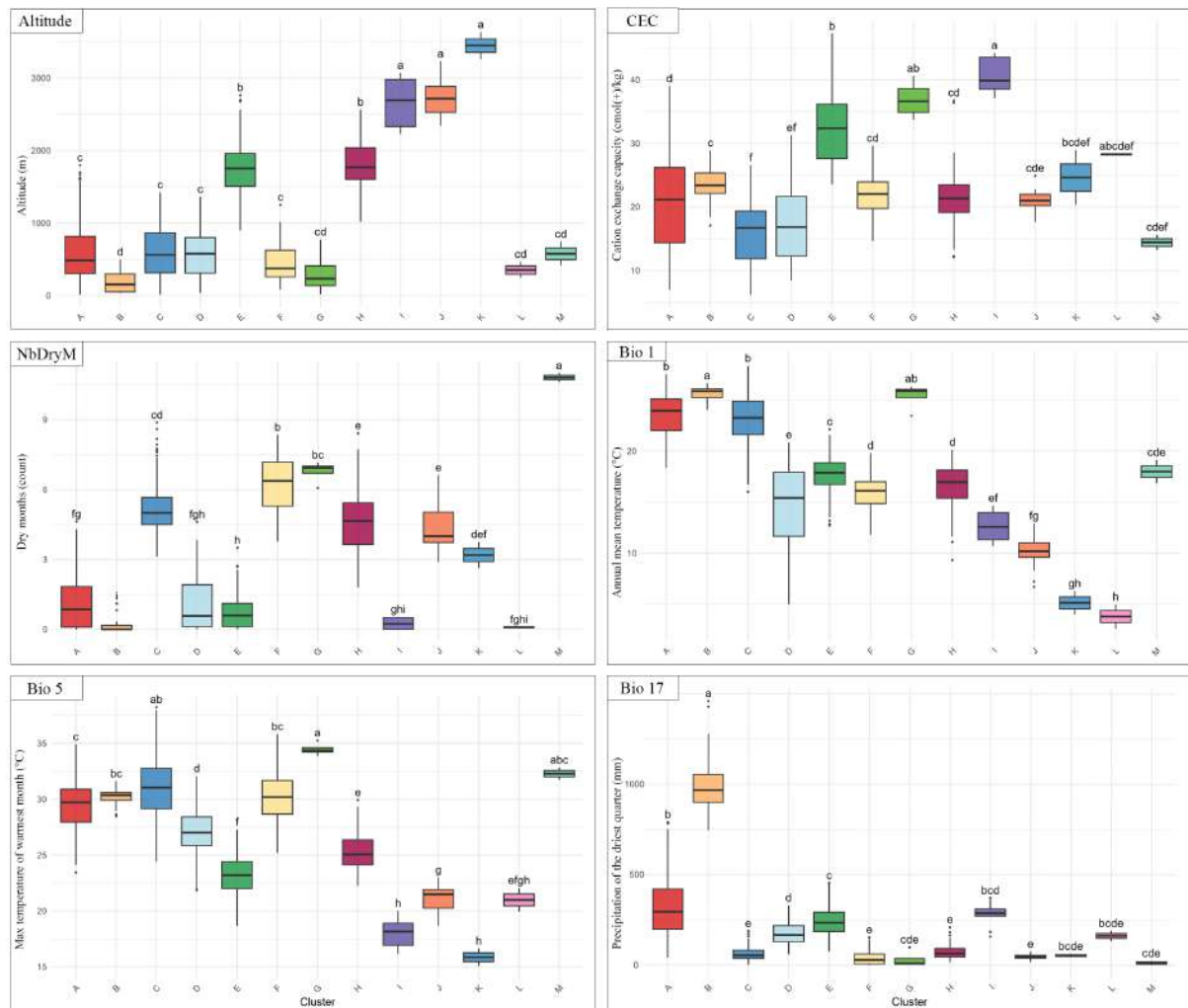


Figure 3: Distribution of environmental variable values across the 13 clusters identified from a principal component analysis (PCA) performed on the mean occurrence values of Araceae species for the following variables: annual mean temperature (Bio1), maximum temperature of the warmest month (Bio5), precipitation of the driest quarter (Bio17), elevation, number of dry months (NbDryM), and soil cation exchange capacity (CEC). Colors correspond to the clusters defining the identified ecological niches. Differences among groups were tested using a one-way analysis of variance (ANOVA), followed by Tukey's honestly significant difference (HSD) post-hoc test (P -value < 0.001), with significance letters assigned based on multiple comparisons.

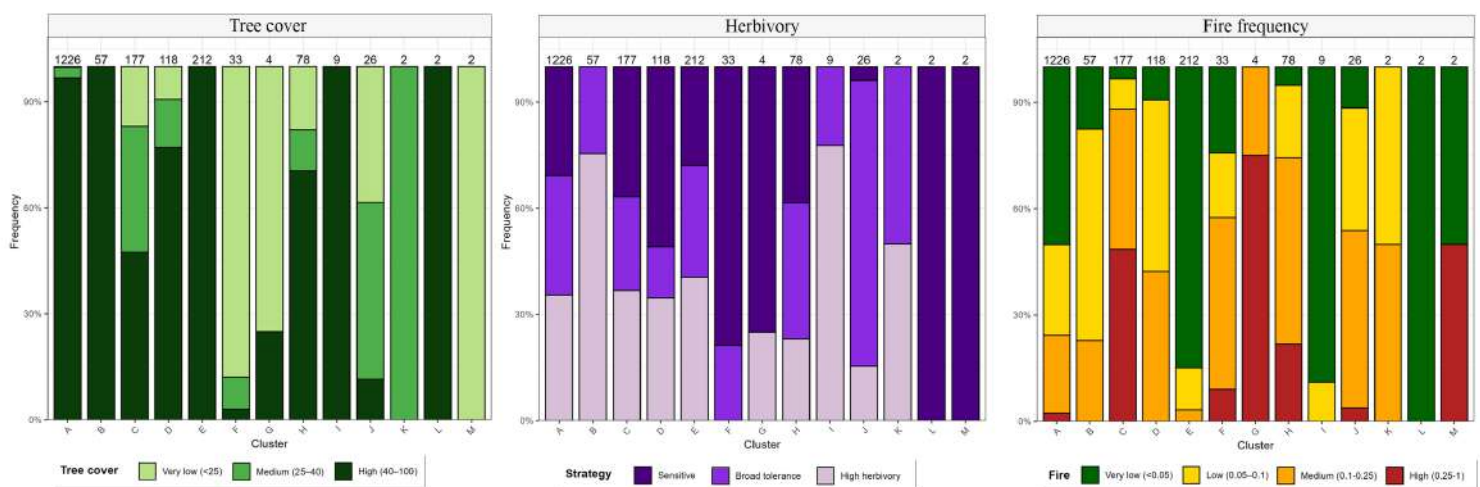


Figure 4: Distribution of frequency values (in %) for tree cover, herbivore pressure, and fire disturbance across the 13 ecological niche clusters. For each variable, occurrences were categorized into classes and their relative frequencies were calculated for each cluster. The total number of species per cluster is indicated above each bar.

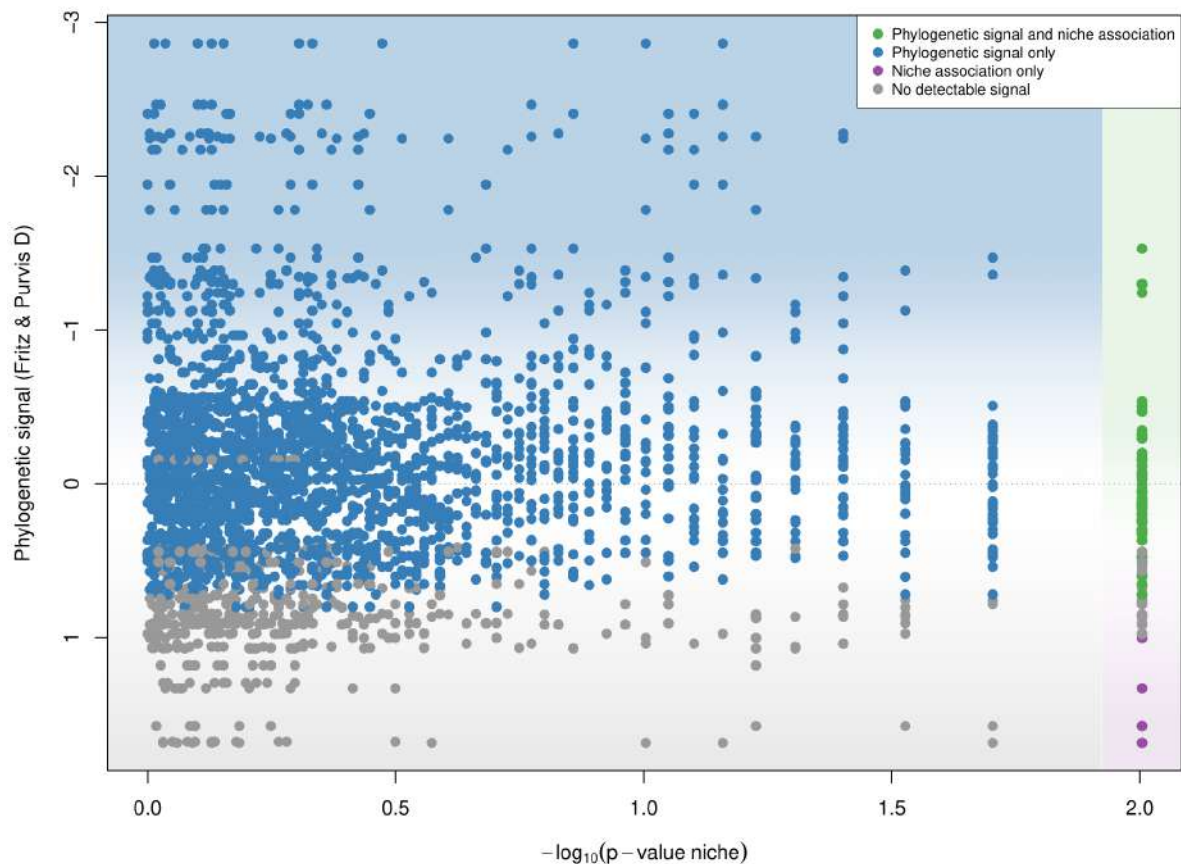


Figure 5: Relationship between phylogenetic signal (Fritz and Purvis’ D statistic) and niche association strength ($-\log_{10}$ -transformed p -values obtained from phylogenetic ANOVA using 1000 permutation-based null models) across 84 binary traits in Araceae genera. Each point represents a trait–niche modality combination after alignment of trait data, niche occurrence matrix, and the pruned phylogenetic tree. Trait data were compiled from Cusimano et al. (2011) and Daniel et al. (In review), phylogenetic signal was calculated following Haigh et al. (2022), and niche associations were tested independently for each trait–niche pair. Points are categorized according to the method of Anest et al. (2024), based on the combined significance of phylogenetic signal and niche association: KC, both significant; CD, phylogenetic signal only; C, niche association only; N, no significant signal.

3. Categorisation of traits and life forms

The combined analysis of phylogenetic signal and niche association identified 42 traits (KC) with both an association to a specific niche and a strong phylogenetic signal, thus meeting the criteria of the KC category ($D < 1$, p_{signal} and $p_{\text{niche}} < 0.01$) (Figure 5). Life form was the trait showing the highest recurrence of niche association across the results, both in the dataset of Cusimano et al. (2010) and in Daniel et al. (In review), emerging as a KC-category trait for six distinct niches. Among the 84 variables examined, the “geophyte” life form was the most recurrent and was positively associated with niches A, F, J, and M (Figure 6). In each of these niches, the trait displayed a D value of 0.319, indicating a strong phylogenetic signal, together with very low ANOVA p -values. The related anatomical trait “tuberous” showed a similar and even more recurrent pattern, being classified as KC for five niches (A, G, H, J, and M) with a phylogenetic signal of $D = 0.297$. The “rhizomatous” life form complemented these underground strategies, being associated with three niches (B, C, and H) with $D = 0.236$. Consequently, life form traits were retained for ancestral state reconstruction analyses using the most recent and robust dataset from Daniel et al. (In review).

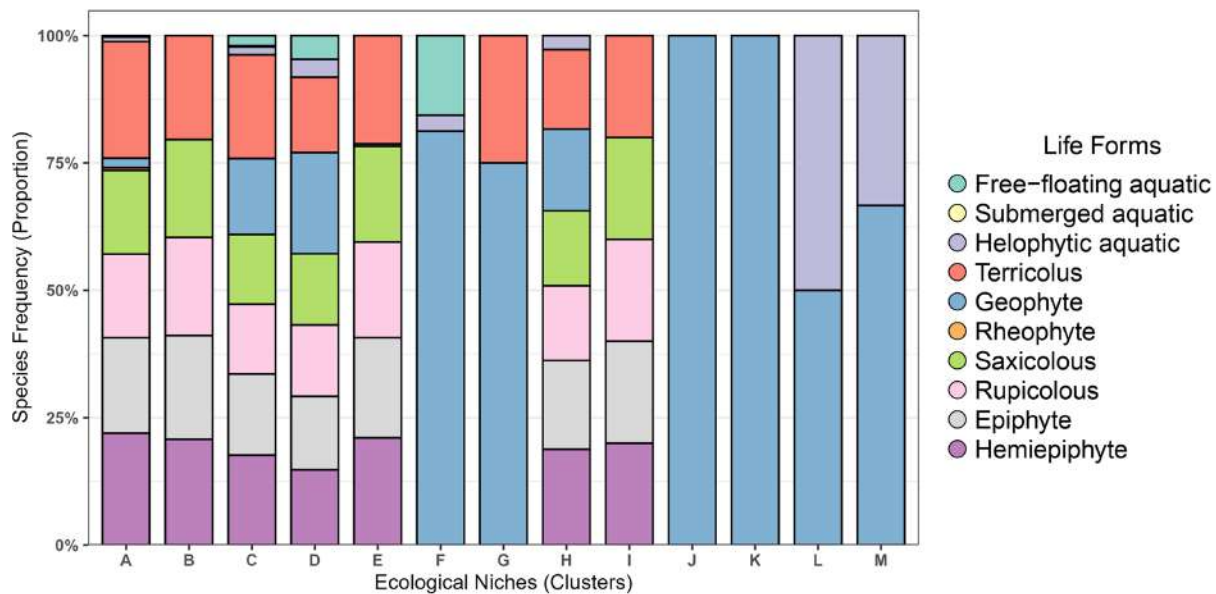


Figure 6: Proportion of Araceae species assigned to each ecological life form, free-floating aquatic, submerged aquatic, helophytic aquatic, terrestrial, geophyte, rheophyte, saxicolous, rupicolous, epiphyte, and hemiepiphyte, across 13 ecological niches (A–M), based on trait data from Daniel et al. (In review)

4. Ancestral states of life form

For ancestral state reconstruction of life forms, the best-fitting evolutionary scenario corresponded to the symmetric transition model (SYM; $AIC_c = 391.94$; Table 2, Appendix). Reconstructions revealed relatively similar probabilities among life forms for the common ancestor of the Araceae, although the helophytic state showed a slightly higher marginal probability (13.7%) (Figure 7).

The ancestral node of the Araceae (~ 128 Mya) displayed relatively elevated probabilities for the aquatic helophytic state (13.7%), terrestrial state (10.1%), and hemi-epiphytic state (11.8%), suggesting substantial ancestral ambiguity at the base of the family (Figure 7). This uncertainty persisted within the Proto-Araceae clade, where the aquatic helophytic state remained slightly dominant (26.4%, ~ 95 Mya). The free-floating state was inferred at the node of *Spirodelia* (63.8%, ~ 80 Mya), leading to near-complete support for the aquatic free-floating condition within the Lemnoideae.

The helophytic state have higher support in ancestral nodes from ~ 110 Mya onward, reaching probabilities of 77.4% at the *Lysichiton* node (~ 70 Mya) and 91.4% at the *Symplocarpus* node (~ 50 Mya), and was characteristic of the Lasioideae (82.9%, ~ 70 Mya) (Figure 7). In contrast, the geophytic state emerged as the best-supported ancestral condition replacing the helophytic state across the majority of Araceae lineages. The probability of the geophytic form is higher from ancestral nodes within the Zamioculcadoideae (Geophyte: 35.5%; Helophyte: 39%; ~ 80 Mya), reaching 94% at ~ 40 Mya, and within the Aroideae (Geophyte: 22.8%; Helophyte: 62%; ~ 77 Mya), eventually attaining 56.4% around 40 Mya (versus 31.7% for the helophytic state) (Figure 7).

The geophytic condition also characterized highly diversified clades including *Pistia* clade and *Arisaema* clade, whose common ancestor was inferred as geophytic around 30 Mya (Figure 7). A recent transition toward the terrestrial state (~ 10 Mya) was inferred within the *Philodendron* clade, embedded in a predominantly terrestrial lineage. A distinct clade underwent a transition toward the rheophytic state from ~ 20 Mya onward, with high ancestral probabilities for this condition (85.4%) across 23 genera, reaching near-

5. Reconstruction of ecological niches and continental ranges

Based on trait selection, we selected life form states as a potential multiplier on dispersal rates for geophytes, aquatics (helophyte + submerged + free-floating) and epiphytes (epiphytic + hemi-epiphytic). We tested the same states effect on dispersal rates for geographical ranges, as a null hypothesis given that vegetative structure should not be particularly associated with long distance dispersal. Because multiple ecological states were accepted per genus, the reconstructions involve substantial uncertainty, particularly for deeper nodes. Several broad patterns can nonetheless be identified.

Details of model selection results for ecological niches and continental ranges are provided in the Appendices (Figure 8, 9). For the model selection for ancestral niche inference including life-form traits-geophytic (Table 5, Appendix) and aquatic (Table 6, Appendix), the best-fitting models are *BAYAREALIKE* ($AIC_c = 718.39$ and 671.19 , respectively), without inclusion of the trait-dependent parameter m_2 . This indicates that accounting for these forms does not improve the fit of the models. The same model is retrieved for the reconstruction of geographical ranges with geophytes ($AIC_c = 680.30$), aquatic ($AIC_c = 633.10$), and epiphytes ($AIC_c = 598.06$) (Table 3, Appendix). In contrast, for epiphytic traits, the best-fitting model for ancestral niche inference is *BAYAREALIKE* + m_2 ($AIC_c = 630.67$) (Table 7, Appendix). The model highlights an asymmetry in dispersal rate multipliers between trait states, with a parameter m_1 estimated at 0.367 (non-epiphytic state) and m_2 fixed at 1 (epiphytic state). This structure indicates a reduction in the dispersal rate associated with non-epiphytes compared to epiphytes, suggesting a facilitating role of the epiphytic life form in ecological transitions.

6. Ecological reconstruction

The common ancestor of the ecological reconstruction is uncertain, which is linked to nodes that are not at the genus scale (Figure 8). Nevertheless, certain groups tend to remain within niche A; this is the case for the Schismatoglottideae, Lasioideae, and Caladieae clades. After this initial occupation, other lineages seem to have diversified their habitats; the Areae appear to have occupied niches in extreme regions for the Araceae (niche R), originating around ~ 10 Mya, but current representatives occupy numerous other niches (C, D, E, H). The *Philodendron* clade follows this pattern, with an origin that seems associated with niche A around ~ 10 Mya, but whose extant representatives are present in C, E, and R. The Proto-Araceae are found today almost exclusively in niche D, which is highly represented among the genera of the family. The Zamiodulcadoideae are present uniquely in niche C.

7. Geographical reconstruction

The reconstructed common ancestor on the continents exhibits an uncertain geographic origin, particularly due to the broad distribution already observed in basal lineages (Figure 9). Very few groups are restricted to specific geographical zones, but some, such as the Schismatoglottideae, appear to have remained in Eurasia after colonizing it 20 Mya ago. Similarly, the Spathicarpeae have remained in America since around ~ 25 Mya. *Wolffia* and its sister groups occupy the four major continental regions considered, whereas the Areae are mainly distributed between Africa and Eurasia, and the Caladieae between America and Eurasia. Conversely, other lineages display more heterogeneous configurations. This is notably the case for *Philodendron*, which groups genera displaying varied biogeographical combinations from Africa-America

to America-Eurasia-Australia.

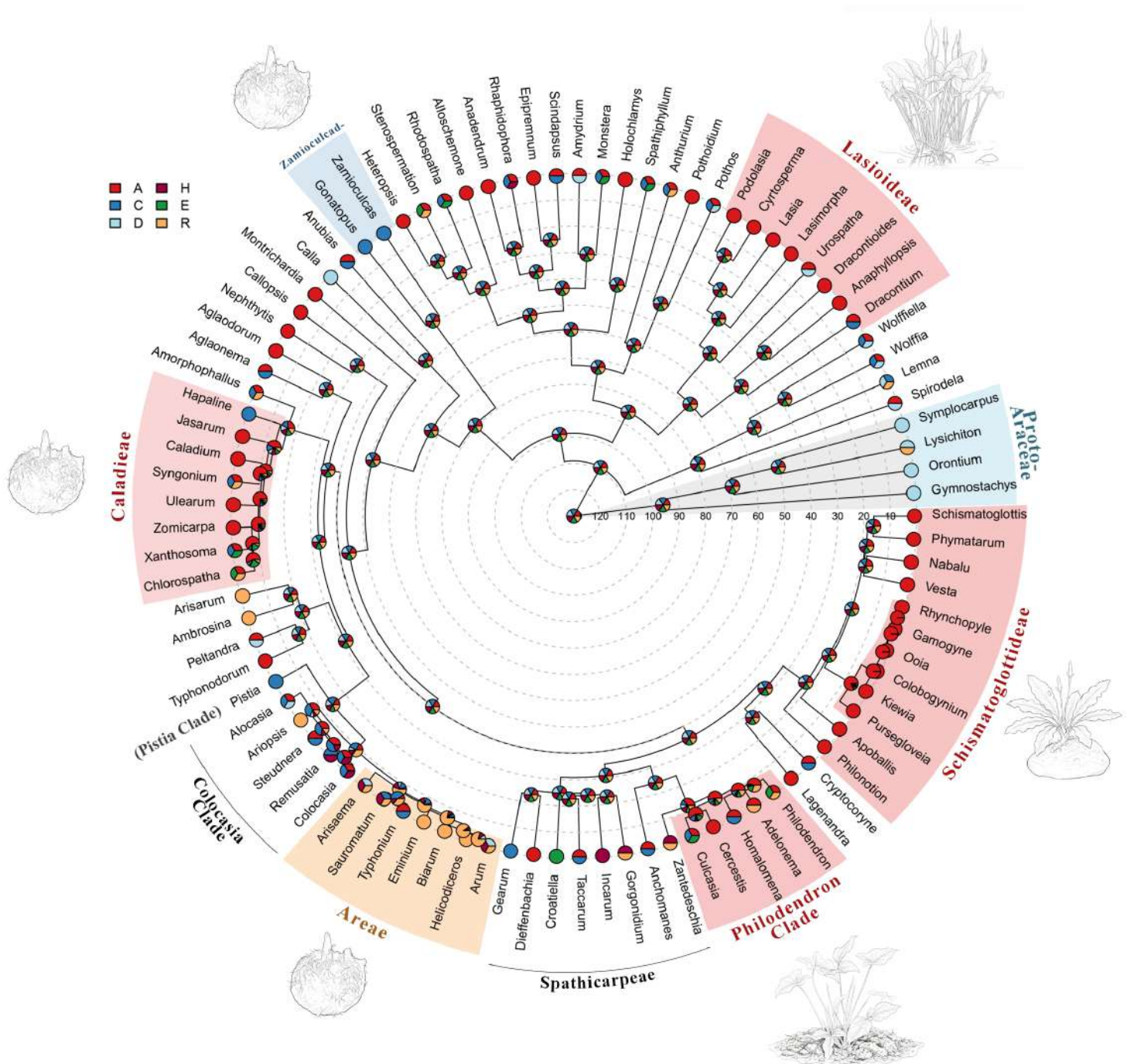


Figure 8: Ancestral ecological niche range reconstruction for the 93 sampled genera of Araceae. Node probabilities were estimated from the marginal probabilities of niches states inferred by BioGeoBEARS and then converted into area-specific occurrence probabilities by summing the contributions of all states containing each region. Ancestral reconstruction of ecological niches under the best-fitting model *BAYAREALIKE* + (m_2) for the epiphytic life form ($AIC_c = 630.67$) and (m_2). Colored regions correspond to clades currently occupying the same ecological niche. Illustrations represent the life forms associated with each clade under the best-fitting ancestral state reconstruction model. Dotted circles correspond to lineage divergence times estimated using a calibration based on the node between *Symplocarpus* and *Orontium*.

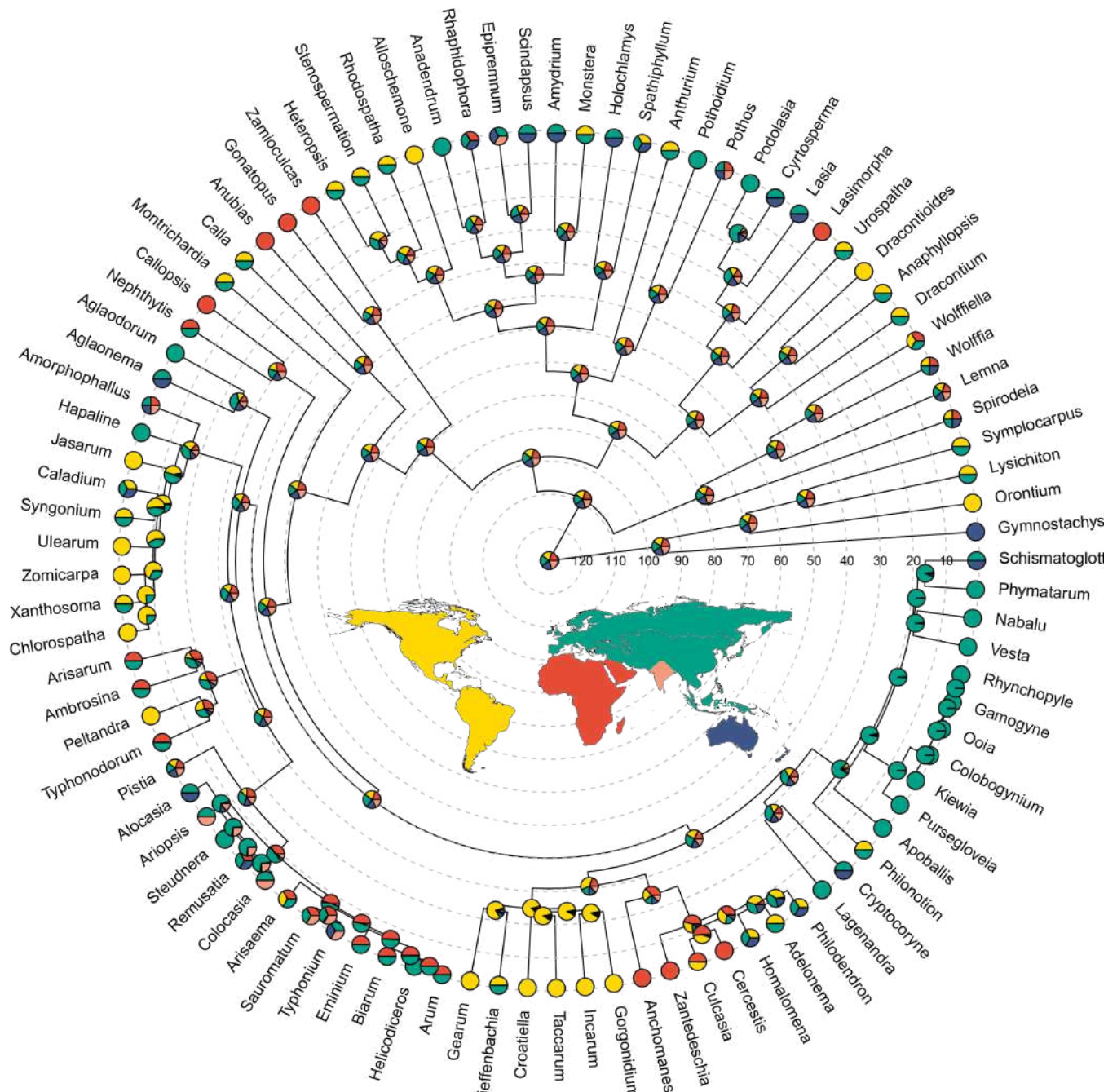


Figure 9: Ancestral geographical range reconstruction for the 93 sampled genera of Araceae. Node probabilities were estimated from the marginal probabilities of geographic states inferred by BioGeoBEARS and then converted into area-specific occurrence probabilities by summing the contributions of all states containing each region. Ancestral reconstruction of ecological niches under the best-fitting model *BAYAREALIKE* for the geophytic life form ($AICc = 680.30$). Dotted circles correspond to lineage divergence times estimated using a calibration based on the node between *Symplocarpus* and *Orontium*.

1.4 Discussion [1740 words]

We tested whether Araceae (1.i) occupy distinct tropical niches identifiable through pedoclimatic characteristics; (1.ii) whether associations exist between occupied niches and life forms; and (2) whether some of these life forms, notably geophytism, have facilitated lineage colonization into seasonal niches. We identified 13 contrasted niches and life forms associated with each of them. Our results do not support the hypothesis that geophytism facilitated ecological diversification, whereas other life forms, such as epiphytism, may have played such a role.

Because our analyses were conducted at the genus level, they incorporate the variation existing among

each genus and therefore provide a macroevolutionary overview rather than a direct test of trait effects at the species level. Reconstructions of ecological niches and geographic ranges were strongly affected by uncertainty associated with genus-level analyses, especially for deep nodes. Despite this limitation, coherent patterns emerge at the family scale, making it possible to identify lineages that should be investigated further using future species-level phylogenies (Daniel *et al.*, in review).

1. Present-day distribution of Araceae

Our study confirms that the majority of *Araceae* occupy the intertropical belt, with eight out of thirteen niches corresponding to tropical, equatorial, or subtropical environments. Our work highlights the existence of more specialized tropical niches shaped by climate, topography, and soil characteristics (Figure 2). The lowland aseasonal equatorial rainforest niche (B), restricted to northwestern Amazonia, corresponds to aseasonal forests characterized by extremely high precipitation levels (Garwood *et al.*, 2023). The high-elevation humid tropical montane forests (I), which are particularly nitrogen-poor, make these ecosystems and their associated forest species highly vulnerable to climatic variation (Gay *et al.*, 2022). This is the case for *Anthurium coerulescens*, currently classified as Critically Endangered (Hermes *et al.*, 2018; IUCN, 2026). Other tropical niches extend across all tropical continents and are occupied by the majority of *Araceae*, whereas basal genera such as the *Zamioculcadoideae* (basal *Aroideae* lineage), *Hapaline* (*Caladieae*), *Pistia* (*Pistia* clade), and *Cryptocoryne* (*Schismatoglottideae*) occupy the humid tropical savanna with seasonal fire regime niche (C). Extreme niches are also represented within *Araceae*, including cold environments (L) and high-elevation Himalayan habitats (K, up to 3,500 m), predominantly occupied by the genus *Arisaema*, which includes critically endangered species associated with these environments (e.g., *Arisaema kawashimae*).

The existence of these specific niches within tropical systems is therefore confirmed by the *Araceae* model, while simultaneously providing new insights into the biogeography of extant genera and shedding light on their past ecology.

2. Evolution of life forms, past ecological niches and continental distributions

Proto-*Araceae* currently occupy the temperate deciduous and mixed forest niche (D), a niche shared by relatively few other *Araceae*. Among the notable exceptions, *Arum* includes species that have shifted into niche D, a pattern also observed in *Arisaema*. The latter represents a particular case: with eight occupied niches in our analysis, it illustrates the limitations inherent to genus-level approaches, where intrageneric diversity tends to blur node state probabilities, while simultaneously making it an ideal model for understanding the processes underlying ecological niche diversification (Tran *et al.*, 2022).

The *Lemnoideae*, characterized by a free-floating life form, are associated with a wide diversity of ecological niches and continental distributions, although the ancestral origin of the clade could not be identified. Our results corroborate those of Crawford *et al.*, 2006, who showed that duckweed speciation is primarily driven by ecological differences rather than strong geographic barriers. Closely related species frequently differ in their ability to tolerate cold, drought, or specific environmental conditions, which may have constituted the initial isolation mechanisms leading to lineage divergence (Figure 8) (Crawford *et al.*, 2006). The presence of *Lemnoideae* fossils in Early Eocene freshwater deposits (~ 55 Ma) indicates that these plants

already occupied continental aquatic environments in regions that are now arid, suggesting a long-standing association of the group with aquatic habitats (Ali *et al.*, 2026).

Our results indicate that Zamioculcadoideae occupy a niche characterized by recurrent fire regimes, exclusively in Africa, and are associated with a geophytic life form. Caladieae diversified in both America and Eurasia within tropical niche A approximately 12 Ma ago, corresponding to a period of climatic cooling following the Middle Miocene Climatic Optimum (Pound *et al.*, 2011). Supported by a geophytic ancestor, it would be interesting to test whether this clade diversified within this niche owing to this life form, which is recognized as an adaptation to highly disturbed environments, particularly those shaped by fire (Lamont *et al.*, 2011; Pausas *et al.*, 2018).

The *Pistia* clade comprises the majority of taxa occupying the most open and arid ecological niches within Araceae (Appendix 6), and is mainly divided into two groups. Their transition from a helophytic to a geophytic life form approximately 40 Ma ago appears to coincide with higher inferred probability of a Eurasian origin for the ancestral lineage of the clade, a signal that becomes stronger in subsequent nodes. The emergence of this strategy may therefore have been associated with selective pressures operating within this new environment, which at that time was dominated by large browsing herbivores (~ 40 Ma), preceding the major faunal restructuring associated with the Eocene-Oligocene transition (~ 34 Ma) (Feijó *et al.*, 2022; Sun *et al.*, 2014).

Monsteroideae and Pothoideae are two closely related groups occupying a broad diversity of ecological niches, yet they remain consistently associated with dense forest habitats, suggesting adaptive strategies responding both to light limitation and to the constraints imposed by the various ecological niches they occupy. The clade originated around 60 Ma, a period during which closed-canopy tropical forests emerged following the K-Pg extinction event, with vertical canopy stratification establishment (Graham *et al.*, 2019). The acquisition of epiphytism around 73 Ma within this group (Figure 7), as suggested by Zhao *et al.*, 2023, combined with its slow diversification until approximately 30 Ma, suggests a gradual exploitation of the different canopy strata. The combination of high ecological niche diversity and low probabilities associated with the ancestral niche of *Monsteroideae* makes the ecological interpretation of this group challenging.

3. Contribution of forms to colonization across ecological niches

The four best ecological niche reconstruction models consistently suggest a multiplicative effect of the epiphytic life form on macroevolutionary dispersal rates within Araceae (Table 7, Appendix), approximately three times greater ($m_2 = 1$) than the estimated parameter for non-epiphytic taxa ($m_1 = 0.37$). This result indicates that (hemi)-epiphytic life forms may have facilitated the colonization of new ecological niches within Araceae. This effect does not reflect physical dispersal ability, but rather an increased capacity for establishment in novel environments, consistent with the hypothesis of Donoghue *et al.*, 2014 that pre-existing “enabling traits” can facilitate biome shifts by determining whether a dispersal event results in successful colonization.

By allowing germination at elevated positions within the canopy while maintaining contact with the ground, the epiphytic strategy may have facilitated establishment in closed-canopy habitats for lineages originating from more open environments, while also promoting transitions into drier habitats through morphological and physiological adaptations to water stress, as demonstrated in hemi-epiphytic *Ficus* species (Zotz *et*

al., [2021]; Zhang *et al.*, [2015]; Gaudry *et al.*, [2026]). Access to the canopy provides a distinct functional space characterized by greater light availability, more intermittent water supply, and unbuffered thermal fluctuations, thereby constituting an environment functionally distinct from the forest floor (Zotz *et al.*, [2021]; Bohlman *et al.*, [1995]; Ingram *et al.*, [1993]). Several studies have shown that epiphytism represents a major ecological innovation promoting diversification in tropical plant lineages, notably in Orchidaceae, Bromeliaceae, and Polypodiaceae (Gentry *et al.*, [1987]; Sundue *et al.*, [2015]; Frenze *et al.*, [2016]), a pattern that remains to be tested in Araceae.

In contrast, geophytic and aquatic models do not show a multiplicative effect on dispersal rates, contrary to our initial hypothesis regarding geophytism (Table 5 and 6, Appendix). Although the best-supported model is *BAYAREALIKE* without m_2 , the second-best model including m_2 presents very similar AICc values despite the difference in parameter number. In this second model, the estimated m_1 parameter for geophytes is greater than 1, suggesting that geophytism may instead reduce the probability of colonizing new niches. This result is consistent with the idea that geophytic life forms, which are strongly associated with recurrent fire regimes, tend to promote strong niche conservatism (Wiens *et al.*, [2005]). By strongly structuring these environments, fire regimes may effectively constrain geophytic lineages within their current habitats and limit transitions toward other ecological niches.

As expected, the aquatic life form does not facilitate the colonization of new niches. Meseguer *et al.*, [2022] demonstrated, using more than 10,000 angiosperm genera, that aquatic lineages exhibit significantly lower net diversification rates than their terrestrial counterparts because of profound structural modifications that constrain transitions toward alternative life forms, effectively constituting an evolutionary dead end in the sense of Day *et al.*, [2016]. Given the probable aquatic origin of the family, diversification of life forms within Araceae may have enabled the group to escape this aquatic “dead end” and colonize the numerous ecological niches currently represented within the family.

4. Implications for the conservation of Araceae

The results of this study reveal two contrasting evolutionary dynamics with direct implications for Araceae conservation. Hemi-epiphytism facilitates establishment in new niches and transitions among habitats, thus constituting a trait that may enhance responses to environmental change, whereas geophytism and aquaticity tend respectively to constrain lineages within their current niches through niche conservatism (Wiens *et al.*, [2005]) or as an evolutionary trap syndrome (Meseguer *et al.*, [2022]; Day *et al.*, [2016]).

Under ongoing climate change and increasing anthropogenic pressures, (hemi-)epiphytism may facilitate niche shifts, as (hemi-)epiphytic species experience contrasting environmental conditions throughout their development, potentially predisposing them to greater ecological plasticity (Donoghue *et al.*, [2014]; Zotz *et al.*, [2021]). This hypothesis is consistent with the life strategy observed in many Monsteroideae, where plants develop reduced-leaf stolons enabling persistence over long distances and extended periods until a suitable support and environment are encountered (Mayo *et al.*, [1997]; Croat, [2019]).

However, (hemi-)epiphytic Araceae are predominantly restricted to closed humid tropical forests and are therefore particularly vulnerable to habitat destruction. Forest fragmentation represents the most thoroughly documented anthropogenic threat to this group: epiphytic species richness may decline by up to 96% in secondary forests relative to primary forests owing to dependence on host trees and specific microclimatic

conditions (Krömer *et al.*, 2025). The conservation of old-life forests and large trees should therefore be considered a priority for these lineages (Nadkarni, 2023).

For geophytes, the major anthropogenic threat lies in policies promoting complete fire suppression, which have been widely adopted across many tropical regions. Natural fire regimes are essential for maintaining savanna biodiversity (Scott, 2000; Bond *et al.*, 2005), and our results suggest that geophytic *Araceae* lineages, which are highly specialized to these environments and exhibit a low probability of colonizing alternative niches, would be particularly vulnerable to the disappearance of such fire regimes. Although the increase in fire frequency predicted for seasonal environments and tropical savannas (Durigan *et al.*, 2020) may temporarily benefit these lineages relative to non-geophytic species occupying the same habitats, both uncontrolled fire intensification and complete fire suppression represent equally unfavorable long-term scenarios. Controlled burning therefore appears to be the most appropriate conservation strategy for these lineages (Bond *et al.*, 2005), maintaining the disturbance regimes to which they are adapted without exacerbating their intensity.

These findings highlight the value of macroevolutionary indicators as complementary tools for identifying intrinsically vulnerable lineages. Applied at the species level and combined with diversification such indicators would allow more refined vulnerability assessments and the development of an integrated framework incorporating predicted climatic changes and current anthropogenic pressures, thereby providing a novel basis for prioritizing conservation efforts within *Araceae*.

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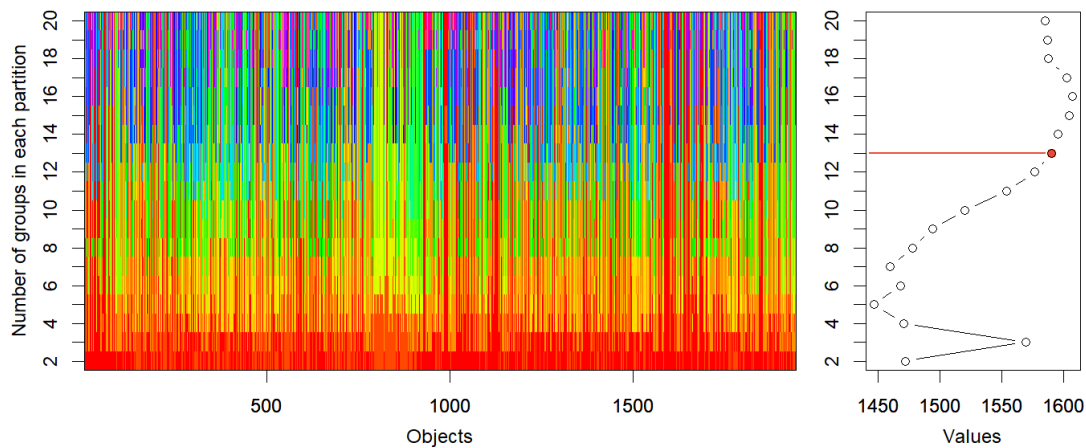
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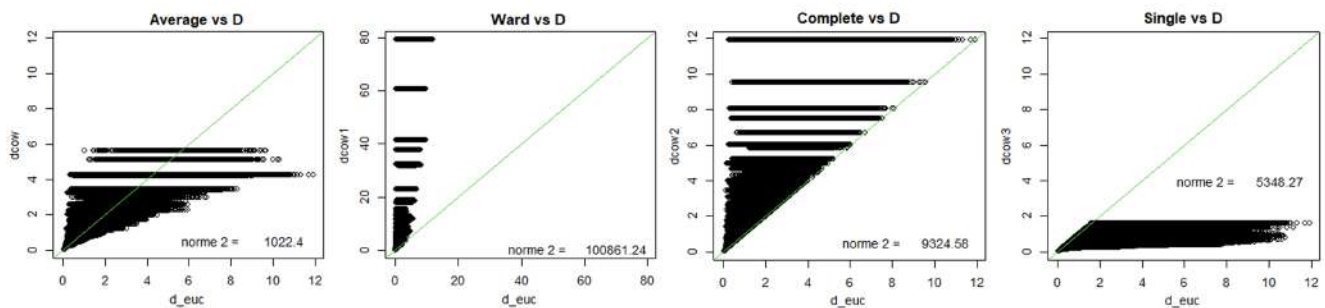
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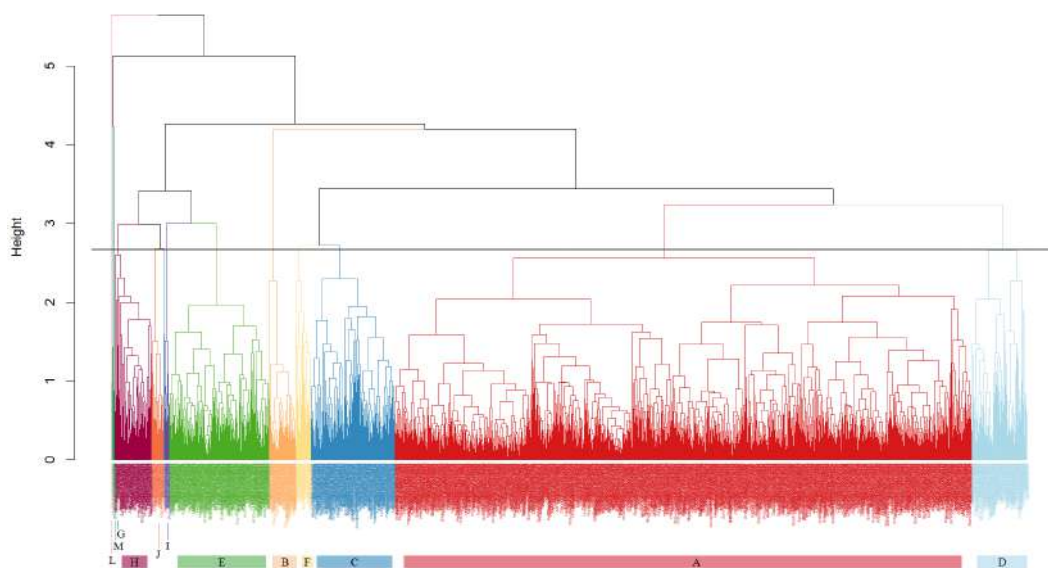
1.5 APPENDICES



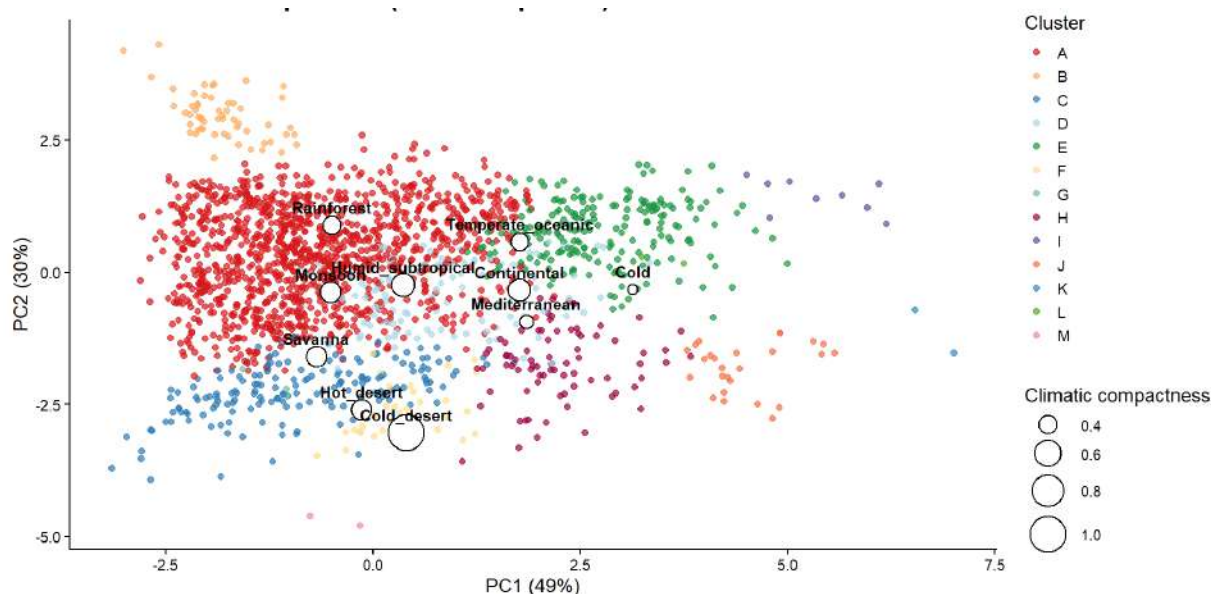
Appendix 1: Comparison of K-means partitions for different numbers of clusters (left) and evaluation of partition quality using the Calinski-Harabasz criterion (right). Each vertical bar represents an object, and colors indicate group assignments. The red tips shows the number of groups retained ($k = 13$), indicating that this partition provides the best balance between within-cluster compactness and between-cluster separation among the tested solutions



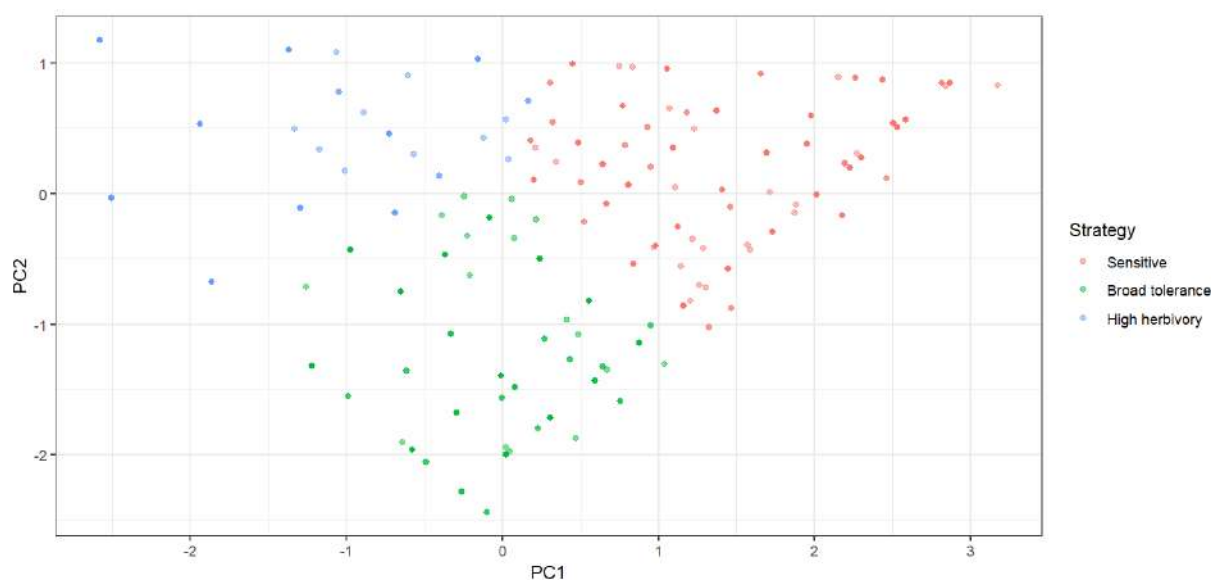
Appendix 2: Euclidean dissimilarity (d_{euc}) as a function of cophenetic distance ($dcow$) according to aggregation criteria: average linkage, U-Ward linkage, complete linkage, single linkage. The Euclidean norm values are given for each. Norm-2 threshold value: $\Sigma = 128.28$. Method from Mériqot *et al.*, [2010](#)



Appendix 3: Euclidean dissimilarity represented by a hierarchical clustering method (dendrogram) for 1,981 Araceae species based on six environmental variables: annual mean temperature (Bio1), maximum temperature of the warmest month (Bio5), precipitation of the driest quarter (Bio17), elevation (Alt), number of dry months (NbDryM), and soil cation exchange capacity (CEC). Colors correspond to clusters delineating the identified ecological niches (A–M). Leaves indicate species names. The optimal aggregation method used is average linkage (Appendix 2). The selected partition into 13 clusters is indicated by a horizontal black bar.



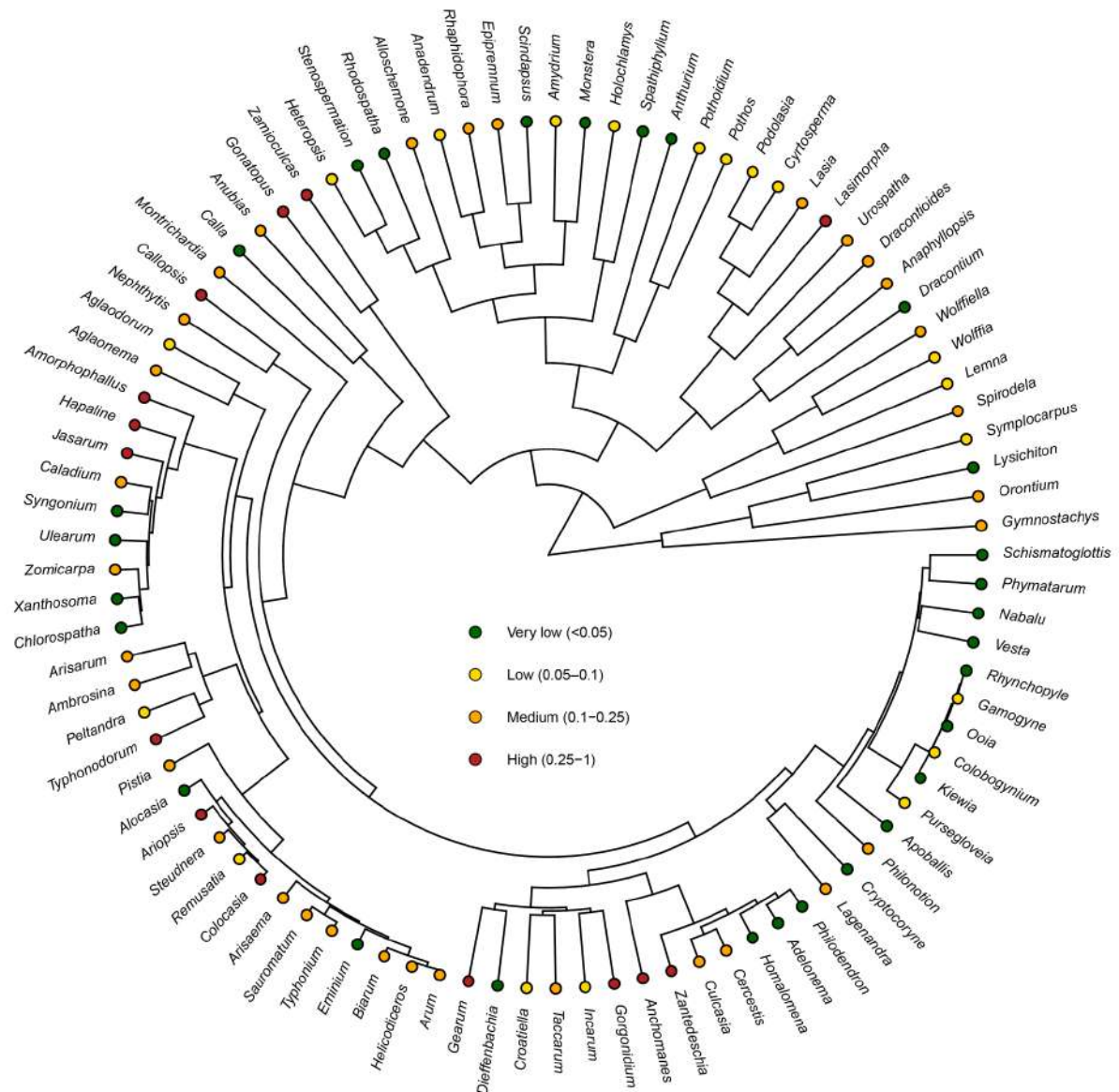
Appendix 4: Passive projection of Köppen–Geiger climatic biome barycentres (Beck et al., 2018) onto the principal component analysis (PCA) space, performed to interpret and refine the characterization of the 13 clusters (A–M) identified as ecological niches. Climatic compactness for each biome is represented as an inverse measure of within-group dispersion in the PCA space, where dispersion corresponds to the mean Euclidean distance of observations to the centroid of each Köppen group, transformed according to the function $\text{compactness} = 1 / (1 + \text{dispersion})$.



Appendix 5: Species-level herbivory strategies inferred from minimum and maximum herbivory values per species. A principal component analysis (PCA) was performed on scaled herbivory minima and maxima, followed by k-means clustering ($k = 3$) to identify dominant herbivory strategies. Clusters were biologically interpreted based on their centroids as cluster : “Sensitive” (low minimum and maximum herbivory), “Broad tolerance” (low minimum but high maximum herbivory) and “High herbivory” (consistently high herbivory). Points represent species projected onto the PCA space, colored according to their assigned strategy.

Table 1: Comparison of life form classifications between Cusimano et al. (2010) and Daniel et al. (in review)

Authors	Life form states
Cusimano et al. (2010)	Rhizomatous, Terrestrial, Caulescent, Climbing, Tuberous, Free-floating.
Daniel et al. (in review)	Free-floating, Submerged aquatic, Helophytic aquatic, Terrestrial, Rheophyte, Rupicolous, Saxicolous, Hemiepiphyte, Epiphyte.



Appendix 6: Genus-level phylogeny of Araceae colored according to fire regimes. Fire regimes were derived from the AVHRR-LTDR ESA CCI Fire product (Chuvieco *et al.*, 2020), covering the period 1982–2018. Annual fire frequency was computed from monthly presence/absence data and subsequently aggregated at the annual scale.

Table 2: Selection of best-fitting evolutionary models for ancestral state reconstruction of life form traits using AIC (phytools; based on data from Daniel et al., in review)

Model	logLik	k	n	AIC	AICc
SYM	-150.9696	45	126	391.9393	443.6893
ER	-203.8948	1	126	409.7897	409.8219
ARD	-135.9036	90	126	451.8073	919.8073

Table 3: Model selection for ancestral inference of the geographic areas. Due to space limitations, the full set of models, along with their outputs and comparisons, is not presented here and is available on the associated GitHub repository.

Life form	Geophyte	Aquatic	Epiphyte
Best model	BAYAREALIKE	BAYAREALIKE	BAYAREALIKE
AICc	680.30	633.10	598.06

Table 4: Phylogenetic signal (Fritz and Purvis' D index) and niche association statistics for binary trait modalities across Araceae genera, reporting trait identity, modality, associated niche, estimated D values, p-values derived from phylogenetic ANOVA using permutation-based null models, and categorical classification of each trait–niche combination, where categories correspond to combined significance of phylogenetic signal and niche association (KC: both significant, CD: phylogenetic signal only, C: niche association only, N: no significant signal).

Traits	States	Niche	D	pval	pval_raw	log_p_niche	categorie
Life.form_mo	Terricolus	E	0,603	0	0,0099	2	KC
Life.form_mo	Terricolus	B	0,603	0	0,0099	2	KC
Life.form_mo	Geophyte	J	0,319	0	0,0099	2	KC
Life.form_mo	Geophyte	M	0,319	0	0,0099	2	KC
Life.form_mo	Geophyte	A	0,319	0	0,0099	2	KC
Life.form_mo	Geophyte	F	0,319	0	0,0099	2	KC
Life.form2_bi	Tuberous	J	0,297	0	0,0099	2	KC
Life.form2_bi	Tuberous	M	0,297	0	0,0099	2	KC
Life.form2_bi	Tuberous	A	0,297	0	0,0099	2	KC
Life.form2_bi	Tuberous	G	0,297	0	0,0099	2	KC
Life.form2_bi	Tuberous	H	0,297	0	0,0099	2	KC
Life.form2_bi	Rhizomatous	B	0,236	0	0,0099	2	KC
Life.form2_bi	Rhizomatous	H	0,236	0	0,0099	2	KC
Life.form2_bi	Rhizomatous	C	0,236	0	0,0099	2	KC
Life.form2_bi	Climbing	E	-0,198	0	0,0099	2	KC
Life.form2_bi	Climbing	B	-0,198	0	0,0099	2	KC
Life.form_mo	Hemiepiphyte	E	-0,295	0	0,0099	2	KC
Life.form_mo	Hemiepiphyte	B	-0,295	0	0,0099	2	KC

Table 5: Model selection for ancestral inference of the 13 niches, with life form. AICc Weight indicates the weight for the model over all the models. Models are ordered from the highest AICc Weight to the lowest. The m_2 models correspond to trait dependant models with a multiplier on dispersal probability when the life form is in state 'geophytic'

Model	lnL	np	AIC	AICc	AICc_weight	rank
BAYAREALIKE+t12+t21	-357.1262	2	718.2524	718.3858	37.49	1
BAYAREALIKE+j+t12+t21	-357.107	3	720.2139	720.4836	13.13	2
BAYAREALIKE+t12+t21+m2	-357.1254	3	720.2507	720.5204	12.89	3
BAYAREALIKE+x+t12+t21	-357.1262	3	720.2524	720.5221	12.88	4
BAYAREALIKE+j+x+t12+t21+m2	-355.8662	5	721.7323	722.422	4.98	5
BAYAREALIKE+j+x+t12+t21	-357.1063	4	722.2126	722.6671	4.41	6
BAYAREALIKE+j+t12+t21+m2	-357.1064	4	722.2127	722.6673	4.41	7
BAYAREALIKE+x+t12+t21+m2	-357.1255	4	722.251	722.7056	4.32	8
DEC+t12+t21	-359.9984	2	723.9969	724.1302	2.12	9
DEC+t12+t21+m2	-359.9716	3	725.9432	726.2128	0.75	10
DEC+x+t12+t21	-360.0012	3	726.0025	726.2721	0.73	11
DEC+j+t12+t21	-360.0081	3	726.0161	726.2858	0.72	12
DEC+j+t12+t21+m2	-359.9805	4	727.961	728.4155	0.25	13
DEC+j+x+t12+t21	-360.0078	4	728.0155	728.4701	0.24	14
DIVALIKE+t12+t21	-362.4528	2	728.9057	729.039	0.18	15
DEC+x+t12+t21+m2	-360.3492	4	728.6984	729.153	0.17	16
DIVALIKE+t12+t21+m2	-362.4306	3	730.8613	731.131	0.06	17
DIVALIKE+x+t12+t21	-362.453	3	730.9059	731.1756	0.06	18
DIVALIKE+j+t12+t21	-362.4605	3	730.921	731.1907	0.06	19
DEC+j+x+t12+t21+m2	-360.312	5	730.6241	731.3137	0.06	20
DIVALIKE+x+t12+t21+m2	-362.4304	4	732.8609	733.3154	0.02	21
DIVALIKE+j+t12+t21+m2	-362.4384	4	732.8769	733.3144	0.02	22
DIVALIKE+j+x+t12+t21+m2	-362.4605	4	732.9211	733.3756	0.02	23
DIVALIKE+j+x+t12+t21+m2	-362.4389	5	734.8778	735.5675	0.01	24

Table 6: Model selection for ancestral inference of the 13 niches, with life form. AICc Weight indicates the weight for the model over all the models. Models are ordered from the highest AICc Weight to the lowest. The m_2 models correspond to trait dependant models with a multiplier on dispersal probability when the life form is in state ‘aquatic’.

Model	lnL	np	AIC	AICc	AICc_weight	rank
BAYAREALIKE+t12+t21	-333.5269	2	671.0537	671.187	32.85	1
BAYAREALIKE+j+t12+t21	-333.0019	3	672.0038	672.2735	19.08	2
BAYAREALIKE+t12+t21+m2	-333.5009	3	673.0018	673.2715	11.58	3
BAYAREALIKE+x+t12+t21	-333.5265	3	673.053	673.3227	11.29	4
BAYAREALIKE+j+t12+t21+m2	-332.9209	4	673.8418	674.2964	6.94	5
BAYAREALIKE+j+x+t12+t21	-333.0018	4	674.0036	674.4581	6.4	6
BAYAREALIKE+x+t12+t21+m2	-333.4955	4	674.991	675.4455	3.91	7
BAYAREALIKE+j+x+t12+t21+m2	-332.9208	5	675.8415	676.5312	2.27	8
DEC+t12+t21	-336.4002	2	676.8004	676.9338	1.86	9
DEC+t12+t21+m2	-335.7945	3	677.589	677.8587	1.17	10
DEC+x+t12+t21	-336.399	3	678.798	679.0677	0.64	11
DEC+j+t12+t21	-336.4084	3	678.8169	679.0865	0.63	12
DEC+j+t12+t21+m2	-335.8349	4	679.6698	680.1243	0.38	13
DEC+x+t12+t21+m2	-336.2185	4	680.437	680.8915	0.26	14
DEC+j+x+t12+t21	-336.4082	4	680.8164	681.2709	0.21	15
DIVALIKE+t12+t21	-338.8533	2	681.7065	681.8398	0.16	16
DIVALIKE+t12+t21+m2	-338.3292	3	682.6584	682.9281	0.09	17
DEC+j+x+t12+t21+m2	-336.1993	5	682.3986	683.0883	0.09	18
DIVALIKE+x+t12+t21	-338.8533	3	683.7065	683.9762	0.05	19
DIVALIKE+j+t12+t21	-338.8609	3	683.7219	683.9916	0.05	20
DIVALIKE+x+t12+t21+m2	-338.3292	4	684.6585	685.113	0.03	21
DIVALIKE+j+t12+t21+m2	-338.3397	4	684.6794	685.1339	0.03	22
DIVALIKE+j+x+t12+t21	-338.8609	4	685.7219	686.1764	0.02	23
DIVALIKE+j+x+t12+t21+m2	-338.342	5	686.684	687.3736	0.01	24

Table 7: Model selection for ancestral inference of the 13 niches, with life form. AICc Weight indicates the weight for the model over all the models. Models are ordered from the highest AICc Weight to the lowest. The m_2 models correspond to trait dependant models with a multiplier on dispersal probability when the life form is in state ‘(hemi)-epiphytic’.

Model	lnL	np	AIC	AICc	AICc_weight	rank
BAYAREALIKE+t12+t21+m2	-312.1987	3	630.3973	630.667	39.93	1
BAYAREALIKE+j+t12+t21+m2	-311.2641	4	630.5283	630.9828	34.1	2
BAYAREALIKE+x+t12+t21+m2	-312.1973	4	632.3947	632.8492	13.41	3
BAYAREALIKE+j+x+t12+t21+m2	-312.2021	5	634.4043	635.0939	4.37	4
BAYAREALIKE+t12+t21	-316.0093	2	636.0187	636.152	2.57	5
DEC+t12+t21	-315.3822	3	636.7644	637.0341	1.65	6
BAYAREALIKE+j+t12+t21	-315.9781	3	637.9563	638.2259	0.91	7
BAYAREALIKE+x+t12+t21	-316.0093	3	638.0187	638.2883	0.88	8
DEC+j+t12+t21	-315.3996	4	638.7991	639.2536	0.55	9
DEC+x+t12+t21	-315.4026	4	638.8051	639.2597	0.54	10
BAYAREALIKE+j+x+t12+t21	-315.9785	4	639.957	640.4115	0.31	11
DEC+j+x+t12+t21	-315.4035	5	640.807	641.4966	0.18	12
DIVALIKE+t12+t21	-317.6262	3	641.2523	641.522	0.18	13
DEC+t12+t21	-318.8821	2	641.7641	641.8975	0.15	14
DIVALIKE+x+t12+t21	-317.6269	4	643.2537	643.7083	0.06	15
DIVALIKE+j+t12+t21	-317.6292	4	643.2584	643.7129	0.06	16
DEC+j+t12+t21+m2	-318.8913	3	643.7826	644.0523	0.05	17
DEC+x+t12+t21+m2	-318.8945	3	643.7891	644.0587	0.05	18
DIVALIKE+j+x+t12+t21	-317.6288	5	645.2577	645.9473	0.02	19
DEC+j+x+t12+t21+m2	-318.891	4	645.7821	646.2366	0.02	20
DIVALIKE+t12+t21	-321.3361	2	646.6721	646.8055	0.01	21
DIVALIKE+x+t12+t21	-321.3361	3	648.6721	648.9418	0	22
DIVALIKE+j+t12+t21	-321.3438	3	648.6875	648.9572	0	23
DIVALIKE+j+x+t12+t21	-321.3438	4	650.6875	651.1421	0	24

1.6 ABSTRACT ENGLISH

Abstract

Background: Traditional extinction risk assessments rely on short-term demographic and spatial data, offering a static view that overlooks macroevolutionary processes and inherited functional constraints. This study aims to infer the macroevolutionary and ecological history of the highly diverse Araceae family to determine how the integration of evolutionary processes can optimize biodiversity vulnerability assessments.

Methods: Based on a dataset of 338,611 occurrences covering 1,981 species and 110 genera, 13 ecological niches were identified by integrating pedoclimatic and ecological variables. The phylogenetic signal of traits was assessed, and trait–niche associations were tested. Ancestral life forms were reconstructed using stochastic character mapping on a genus-level phylogeny, and trait effects on niche and geographical range diversification were modeled using BioGeoBEARS.

Results: Our analyses suggest a helophytic origin for Araceae, followed by transitions toward diverse terrestrial strategies. The epiphytic life form likely facilitated the colonization of novel ecological niches. Conversely, geophytic forms might have prevented lineages from colonizing niches with concentrated ecological constraints, thereby contributing to niche conservatism and ecological specialization within environments subject to recurrent fire regimes.

Conclusion: Macroevolutionary legacies observable through life form syndromes provide promising insights for estimating lineage vulnerability to extinction risk, both independently and in relation to direct anthropogenic pressures. This study highlights that different life forms with different evolutionary histories can be differently affected by future changes and threats. A species-level integrated model is now required to better evaluate lineage vulnerability and prioritize conservation efforts in Araceae.

Keywords: Macroevolution, Biogeography, Plant trait evolution, Conservation, Extinct risk, Araceae

1.7 ABSTRACT FRANÇAIS

Abstract

Contexte : Les évaluations du risque d'extinction reposent sur des données démographiques et spatiales à court terme, offrant une vision statique qui néglige les processus macroévolutifs ainsi que les contraintes fonctionnelles héritées. Cette étude vise à reconstituer l'histoire macroévolutive et écologique de la famille très diversifiée des Araceae afin de déterminer comment l'intégration des processus évolutifs peut améliorer les évaluations de la vulnérabilité de la biodiversité.

Méthodes : À partir d'un jeu de données comprenant 338 611 occurrences couvrant 1 981 espèces et 110 genres, 13 niches écologiques ont été identifiées, intégrant des variables pédoclimatiques et écologiques. Le signal phylogénétique des traits et les associations entre traits et niches ont été évaluées. Les formes de vie ancestrales ont été reconstruites avec une cartographie stochastique des caractères sur une phylogénie des genres, et les effets des traits sur la diversification des niches et des aires géographiques ont été modélisés à l'aide de BioGeoBEARS.

Résultats : Nos analyses suggèrent une origine héliophytique, suivie de transitions vers diverses stratégies terrestres. La forme de croissance épiphyte a probablement facilité la colonisation de nouvelles niches écologiques. En revanche, les formes géophytiques pourraient avoir empêché certaines lignées de coloniser des niches caractérisées par des contraintes écologiques concentrées, contribuant ainsi au conservatisme de niche et à la spécialisation écologique dans des environnements soumis à des régimes de feu récurrents.

Conclusion : Les héritages macroévolutifs observables à travers les syndromes de formes de vie constituent des éléments prometteurs pour estimer la vulnérabilité des lignées au risque d'extinction, indépendamment et en interaction avec les pressions anthropiques directes. Cette étude souligne que différentes formes de croissance, associées à des histoires évolutives distinctes, peuvent être affectées différemment par les changements et menaces à venir. Un modèle intégré à l'échelle des espèces est désormais nécessaire pour mieux évaluer la vulnérabilité des lignées et prioriser les efforts de conservation des Araceae.

Mots-clés : Macroévolution, Biogéographie, Traits, Conservation, Risque d'extinction, Araceae